

**“IMPLEMENTATION
AND TESTING OF A HYBRID RANS-LES (BCM-DES)
MODEL FOR TRANSITIONAL SEPARATED FLOWS
OVER CURVED BLUFF BODIES”**

A Thesis

**Submitted by
ANUPAM VATS**

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ANUPAM VATS

DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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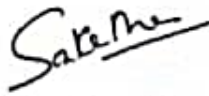
Date: 09/06/26

Name: ANUPAM VATS

Reg. No.: M240984ME

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Dr. Athkuri Sai Saketha Chandra

(Guide)

Asst. Professor

Dept. of Mechanical Engineering



Place : NIT Calicut

Date :



Professor & Head

Dept. of Mechanical Engineering

प्राध्यापक एवं विभागाध्यक्ष / Professor & Head
यांत्रिक अभियंता विभाग
Department of Mechanical Engineering
राष्ट्रीय प्रौद्योगिकी संस्थान कलिकट
National Institute of Technology Calicut
कलिकट - ६७३६०१, केरल / Calicut - 673601, Kerala



राष्ट्रीय प्रौद्योगिकी संस्थान कलिकट

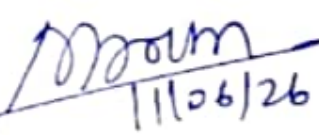
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Department of Mechanical Engineering

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 2. Register number: M240984ME
 3. Title of the project: "IMPLEMENTATION AND TESTING OF A HYBRID RANS-LES (BCM-DES) MODEL FOR TRANSITIONAL SEPARATED FLOWS OVER CURVED BLUFF BODIED".
 4. Date of thesis submission: 09-JUNE-2026
 5. Date of Viva Voce: 11/06/26
 6. Name and Affiliation of External Examiner: Dr. Sathish Kumar D
Associate professor, CED, NITC.

Signature of Guide's/ Programme Coordinator

Name: T. J. Saravanan / hame jhm  11/06/26

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ABSTRACT

Prediction of the aerodynamic forces on bluff bodies or lifting surfaces in transitional flow conditions is still a long-standing problem in the development of computational fluid dynamics. The separation behaviour, pressure distributions and force coefficients of such geometries are ultimately controlled by the nature of the boundary layer (laminar, transitional, or fully turbulent). Most turbulence models, including all the RANS turbulence models, assume that the inlet flow is turbulent and cannot capture the laminar to turbulent transition process. LES is able to simulate turbulent structures at large scales with high fidelity, but it is too costly in terms of computational time to simulate wall-bounded transitional flows at moderate to high Reynolds numbers.

This thesis provides a numerical analysis of transitional separated flows with a novel transition-aware hybrid RANS-LES model developed by coupling Spalart-Allmaras Bas-Cakmakcioglu (SA-BCM) transition model with the Spalart-Allmaras Delayed Detached Eddy Simulation (SA-DDES) framework using a blending parameter, $\Phi = f_d \cdot \gamma_{BC}$. This coupling is designed such that the model is transition sensitive in the near wall region, and only switches to the LES mode in the fully turbulent region of the flow away from the wall. All the simulations are carried out using the in-house compressible unstructured finite-volume solver PRAVAHA-IITH.

Three main test situations are analyzed. First, flow was examined over a stationary circular cylinder between $Re = 5 \times 10^4$ and 10^6 , where the BCM-DES model is able to capture the drag crisis, a sharp drop in drag coefficient caused by Reynolds-number-induced boundary layer transition. Second, flow over a rotating circular cylinder at $Re = 152,000$ at several different values of the spin ratio $\alpha = 0$ to 1.0 , where lift crisis due to rotation-induced transition is observed. The correct capture of the negative lift coefficient at $\alpha = 0.25$, as obtained by the BCM-DES model is also not reproduced qualitatively by the standard DES model. Third, flow over the

Eppler 387 low Reynolds number airfoil, which is accurately captured for laminar separation bubble formation, bubble bursting and static stall.

Validated against flat-plate transition benchmarks, the blending function are found to behave as expected and the level of accuracy in the prediction of the transition is confirmed. The mesh quality is guaranteed by the grid sensitivity studies with the Pope's turbulent kinetic energy resolution criterion.

The results as a whole suggest that BCM-DES is a computational efficient method for simulating separated flows that were in transition, and yielded results similar to those of full LES for the same fraction of the cost.

BCM-DES has improved prediction compared to DES models and sometimes LES because of its transitional capabilities.

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LIST OF SYMBOLS AND ABBREVIATIONS

C_{b1}, C_{b2}	SA model constants
C_{DES}	DES calibration constant (= 0.65)
C_d	Drag coefficient
C_f	Skin friction coefficient
C_l	Lift coefficient
C_p	Pressure coefficient
C_{pb}	Base pressure coefficient
d	Wall-normal distance
$\tilde{d}$	Modified hybrid length scale (DDES)
$d\Phi$	BCM-DES modified length scale
D	Cylinder diameter or airfoil chord
E	Specific total energy
f_d	DDES shielding function
f_{v1}, f_{v2}, f_w	SA model auxiliary functions
H	Specific total enthalpy
k	Turbulent kinetic energy
$\hat{k}$	Effective thermal conductivity
L_z	Spanwise domain extent
L_0	Turbulent integral length scale
Ma	Mach number
n_x, n_y, n_z	Face outward unit normal components
p	Pressure
Pr_L, Pr_T	Laminar and turbulent Prandtl numbers
Re	Reynolds number

Re_v	Vorticity Reynolds number
Re_θ	Local momentum thickness Reynolds number
$Re_{\theta c}$	Critical transition Reynolds number (BCM)
$\tilde{S}$	Modified strain rate in SA model
St	Strouhal number
T	Absolute temperature
Tu_∞	Freestream turbulence intensity
u, v, w	Velocity components
U_∞	Freestream velocity
V	Contravariant velocity
W	Conservative variable vector
α	Spin ratio (rotating cylinder) or angle of attack (airfoil)
β^*	k- ω SST model constant (= 0.09)
γ_{BC}	BCM intermittency function
Γ_Φ	Effective diffusivity
Δ	Local grid length scale (sub grid scale)
Δt	Physical time step
μ_L, μ_T	Laminar and turbulent dynamic viscosity
ν_L, ν_T	Laminar and turbulent kinematic viscosity
$\tilde{\nu}$	Modified SA eddy viscosity variable
Φ	BCM-DES blending parameter ($= f_d \times \gamma_{BC}$)
ρ	Fluid density
τ_{ij}	Viscous stress tensor
Ω	Vorticity magnitude
Ω (rotation)	Cylinder angular velocity
ω	Specific dissipation rate (k- ω SST)
κ	von Kármán constant (= 0.41)
σ	SA model diffusion constant (= 2/3)
χ	Ratio $\frac{\tilde{\nu}}{\nu_L}$ in SA mod

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF TRANSITIONAL FLOWS

The transition from laminar to turbulent flow is often not exactly at the extremes of ideal laminar and turbulent flows, as is often assumed in fluid flow engineering. The boundary layer undergoes a complex transition from a laminar ordered state to a turbulent chaotic state in a wide variety of applications, ranging from aeronautical vehicles to marine structures, turbomachinery, and biomedical devices. This transitional regime, in which laminar, turbulent, and intermittent regions occur simultaneously on a single body, is more difficult to predict than either of the limiting cases.

Transitional flows are those in which the boundary layer transitions from laminar to turbulent flows, which is caused by the local Reynolds number, freestream turbulence intensity, surface roughness, surface curvature, and surface pressure gradient. The transition process involves the growth of Tollmien-Schlichting (T-S) instabilities in the linear stage, development of three-dimensional Λ -vortices, breakdown to turbulent spots, and subsequent merging of the turbulent spots into a fully turbulent boundary layer downstream. At intermediate turbulence intensities, the process is further complicated by mechanisms that bypass the classical T-S route.

From an engineering perspective, the prediction of transitional flow is important because it affects the integrated aerodynamic performance. The pressure distribution, degree of reversed flow, and hence, drag and lift properties of the body depend on the location of the transition to the boundary layer and whether a laminar separation bubble is produced as a result of the transition. Errors in the chord prediction of even a few percent result in errors of more than 50% in the prediction of drag in low-Reynolds-number airfoils.

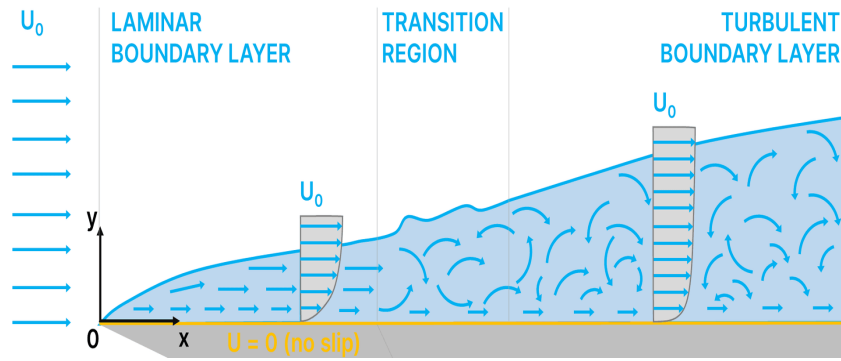


Figure 1.1 shows a schematic of laminar-to-turbulent transition in a flat-plate boundary layer, illustrating stages from linear Tollmien-Schlichting wave growth through vortex formation’.

1.2 IMPORTANCE OF BOUNDARY LAYER TRANSITION

The exchange of momentum in the boundary layer near the wall, and hence the integrated aerodynamic performance of a body, is significantly influenced by the boundary layer state (laminar, transitional, or turbulent). Laminar boundary layer has a small amount of momentum near the wall, and is sensitive to adverse pressure gradients. In contrast, a turbulent boundary layer withstands higher adverse pressure gradients because of increased momentum transport and postpones/separates the flow.

These differences have significant real-world implications. In the case of circular cylinders and spheres, the change from laminar to turbulent separation of the boundary layer typically leads to a decrease in the overall drag coefficient by a factor of 3 to 5 (the so-called drag crisis) because the turbulent boundary layer separates at further downstream locations, drastically decreasing the wake width and the associated pressure drag, which is dominant at these Reynolds numbers. The onset of stalls is dominated by the formation and bursting of a laminar separation bubble for low-Re airfoils at high angles of attack. Initially, the bubble is a small, completely detached recirculating zone that reattaches the separated shear layer to the turbulent boundary layer. Bursting causes the flow to not reattach completely, and a catastrophic stall occurs.

In addition to lift and drag, transition influences the rates of heat transfer (important for turbine blades), acoustic noise generation (important for fan and propeller design), unsteady separated-flow loading (important for structural loading and design of offshore risers), and the occurrence of vortex-induced vibration in offshore risers. Therefore, it is essential to predict the transition correctly, not simply as an academic exercise but as the basis for a reliable engineering design.

1.3 HYBRID RANS-LES MODELLING

The Reynolds-averaged Navier–Stokes (RANS) framework has long served as a computational workhorse for engineering flow simulations. By time- or ensemble-averaging the Navier-Stokes equations, RANS models replace turbulent fluctuations with modelled transport equations for quantities such as eddy viscosity and turbulent kinetic energy. Although computationally economical, RANS models struggle in massively separated flows, bluff body wakes, and configurations with large-scale unsteady vortical structures, characteristically overpredicting the eddy viscosity in separated shear layers and suppressing the natural dynamics of the unsteady wake.

Large Eddy Simulation (LES) directly resolves energy-containing turbulent eddies while modelling only subgrid-scale motions. It provides a much more faithful representation of separated flows, turbulent wake, and transition-related phenomena. However, it imposes severe computational demands in wall-bounded flows at high Reynolds numbers: near-wall mesh spacings must scale as $Re^{-1/3}$ in wall-parallel directions, making wall-resolved large eddy simulation (LES) computationally intractable for practical Reynolds numbers. As a benchmark, a fully wall-resolved LES of a circular cylinder at $Re = 10^6$ would require on the order of 10^9 cells, which is entirely beyond the reach of routine engineering computations.

Hybrid RANS-LES models (mainly Detached Eddy Simulation (DES) and its variants, DDES (Delayed DES) and IDDES (Improved DDES)) are a practical middle ground. RANS is used with relatively coarse meshes for the attached boundary layer in turbulent flow, whereas LES is used for the unsteady separated wake, where computational costs are not as critical. Despite their strengths, all

standard DES variants assume a fully turbulent inflow; they cannot reproduce the laminar-to-turbulent transition process, making them unsuitable for flows where the boundary layer transition plays a central role in the aerodynamic response.

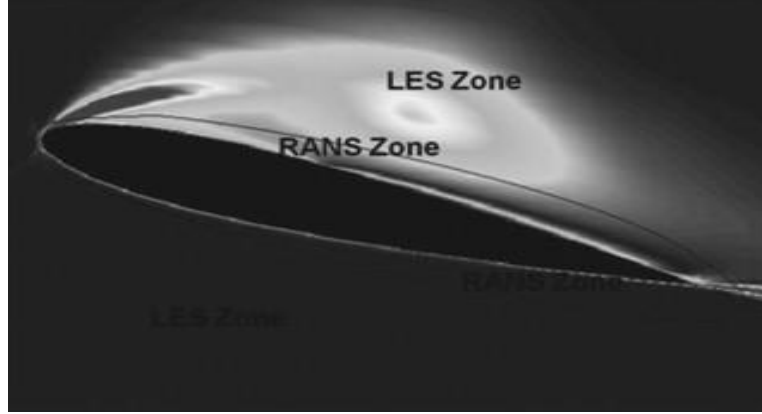


Figure 1.2: Schematic of DES RANS-LES zonal switching around a bluff body, depicting the near-wall RANS region (where preserves wall-distance-based destruction) and separated wake LES region (where activates grid-scale length

1.4 THE BCM-DES MODEL

The Bas-Cakmakcioglu-Mura (BCM) transition model modifies the Spalart-Allmaras (SA) RANS equation using a multiplicative intermittency function that is equal to unity in fully turbulent regions and approaches zero in laminar regions. This function suppresses turbulent production in the pre-transitional boundary layer, allowing the SA equation to remain dormant (laminar) until the local vorticity Reynolds number exceeds the freestream turbulence-dependent threshold. The model is purely algebraic, and no additional transport equations are required, making it computationally negligible.

The BCM-DES model developed in this study combines this transition-detection capability with the scale-resolving power of SA-DDES through a blending parameter, where f_d is the DDES shielding function. This coupling ensures that the LES behavior is activated only when the flow is simultaneously away from the wall ($f_d = 0$) and turbulent ($\gamma_{BC} = 1$), preventing premature LES activation in laminar or transitional boundary layers. The formulation requires no additional transport

equations and can be implemented in any existing DES solver with minimal modifications.

1.5 OBJECTIVES OF THE PRESENT WORK

The primary aim of this thesis is to develop, validate, and apply the BCM-DES transition-aware hybrid RANS-LES model to investigate transitional separated flows over curved bluff bodies and low-Reynolds-number aerodynamic surfaces. The specific objectives are as follows:

Objective 1 - Flow over a stationary circular cylinder: Examine the drag crisis conditions for $Re = 5 \times 10^4$ to 10^6 , where the BCM-DES shows good performance in predicting the onset of boundary layer transition, which causes a dramatic drop in the drag coefficient, and validate its predictions against well-established experimental benchmarks.

Objective 2 - Flow over a rotating circular cylinder: Investigate the lift crisis ($Re = 152,000$) for spin ratios $\alpha = 0, 0.10, 0.20, 0.25, 0.30, 0.40, 0.50$, and 1.0 , resulting in a counterintuitive change in the lift coefficient from positive to negative when the boundary layer transition occurs owing to the rotation effects, an occurrence not captured by the standard DES.

Objective 3 - Flow over the Eppler 387 airfoil: Explore LSB formation, bubble bursting, and static stall at various angles of attack and demonstrate the usefulness of BCM-DES in low-Reynolds-number airfoil aerodynamics, compared to existing NASA Langley experimental data.

1.6 THESIS ORGANIZATION

The second chapter contains an extensive literature survey, which includes experimental and numerical investigations of flow over stationary and rotating cylinders, the Eppler 387 airfoil, transition modelling methods, and hybrid RANS-LES methods, highlighting the areas of research gaps targeted in this study.

The governing compressible Navier-Stokes equations, with the presence of the viscous stress tensor, thermal conductivity, and Sutherland viscosity law, are described in the integral finite-volume form, as detailed in Chapter 3. Chapter 4 presents the PRAVAHA-IITH solver, which is based on a finite-volume

framework, spatial and temporal discretization schemes, the Circle Green-Gauss gradient method, and Pope's turbulent kinetic energy resolution criterion for grid quality assessment.

The turbulence model hierarchy and baseline SA model are detailed in Chapter 5, along with the SA-BCM transition model, newly developed SA-DDES hybrid model, and novel BCM-DES formulation. Chapter 6 validates the BCM-DES model using transition benchmarks of flat plates. The results for the stationary cylinder, rotating cylinder, and Eppler 387 airfoil are presented in Chapters 7, 8, and 9, respectively, with a detailed discussion of the flow physics. The chapter ends with significant findings, contributions, and future directions.

CHAPTER 2: LITERATURE REVIEW

2.1 FLOW OVER CIRCULAR CYLINDERS

The flow over a circular cylinder is one of the most extensively studied flows in fluid mechanics because it is a canonical configuration for the study of bluff body aerodynamics, vortex shedding, and boundary layer transition. The cylinder problem has a rich variety of flow regimes as a function of the Reynolds number, ranging from laminar steady separation at $Re \sim 5 - 40$, periodic Strouhal shedding at $Re \sim 40 - 200$, 3-dimensional wake transition at $Re \sim 190 - 260$, to the fully turbulent regime.

In the subcritical regime ($Re \sim 10^3$ to 3×10^5), the boundary layer on the cylinder surface remains laminar until the separation point, and the shear layers separated from the surface of the cylinder become turbulent in the wake. In this regime, the drag coefficient is relatively independent of the Reynolds number and is of the order $\approx 1.0-1.2$. The most extensive single-study experimental dataset for this geometry is that of Achenbach (1968) [1], who measured the local skin friction and pressure around the cylinder for Re up to 5×10^6 . The velocity and Reynolds stress fields were then established by careful phase-averaged measurements at $Re = 1.4 \times 10^5$ by Cantwell and Coles (1983) [2] using a flying hot-wire technique, which is now a standard set of validation measurements.

The drag crisis, a precipitous drop from ~ 1.0 to below 0.3 , occurs over a range of $Re \sim 3 \times 10^5$ to 5×10^5 . The physical mechanism is that the boundary layer changes from laminar to turbulent separation, the turbulent boundary layer near the wall has higher momentum than the laminar boundary layer, and the separation point is delayed from $\theta \sim 80^\circ$ (laminar) to $\theta \sim 120-140^\circ$ (turbulent), which dramatically reduces the base pressure loss and narrows the wake. The existence of the drag crisis has been classically documented by experiments conducted by Schewe (1983) [3] and Wieselsberger (1922) [4], and was shown by Norberg (1987) [5] to be

accelerated at lower Re by increasing the freestream turbulence, which caused an earlier boundary layer transition.

Travin et al. (2000) [6] found that the drag crisis is not reproduced by standard DES with turbulent inflow since there is no sharp drop in their prediction of between and (for all Re , assuming turbulent inflow in the boundary layer). However, the wall-resolved transition model LES with 38–100 million cells was successful in drag crisis prediction by Rodriguez et al. (2015) [7] and Yeon et al. (2016) [8], showing that transition resolving ability is a necessary requirement. However, these LES approaches are not feasible for engineering applications. The BCM-DES model addresses this issue by employing at least 2.5 million cells.

2.2 FLOW OVER ROTATING CYLINDERS

The Magnus effect (transverse lift force) is generated by the interaction between a rotating cylinder and an incoming freestream. The freestream-relative surface velocities were asymmetric owing to the angular velocity of the cylinder, which caused the Magnus effect. They are used in the propulsion of ships, wind power, and the aerodynamics of cricket and tennis balls. The Magnus effect at high spin ratios is well known, whereas the lift crisis at moderate spin ratios and Reynolds numbers is not as obvious.

Swanson (1961) [9] did extensive experimental studies of the Magnus effect over various Reynolds numbers and spin ratios. At $Re = 152000$ and $\alpha = 0.25$, Case G yielded a negative lift coefficient $C_l = -0.1$, which is contrary to the Magnus expected positive lift for a clockwise rotating cylinder. This is due to the rotationally induced asymmetric boundary layer transition, with greater shear acting on the lower surface boundary layer, causing an earlier turbulent transition, and no shear acting on the upper surface boundary layer to allow the boundary layer to remain laminar for a longer time, according to Swanson [9]. This differential separation angle produces a pressure asymmetry that acts against the classical Magnus direction.

Karabelas (2010) [10] conducted LES at $Re = 140,000$ for spin ratio α between 0 and 2.0 (using ~ 10 million cells). Downstream detailed LES were performed at $Re = 60,000$ at various spin ratios $\alpha = 0-1.5$, with a detailed analysis of the transition mechanism that occurs owing to rotation (Cheng, Pullin and Samtaney, 2018 [11]). Both LES studies showed negative lift for moderate spin ratios, and they were both very demanding in terms of computing resources. The prediction of the monotonically increasing positive Magnus lift with the spin ratio, as given by the standard SA-DDES, is qualitatively incorrect for the moderate- α regime owing to the turbulent inflow on both surfaces. BCM-DES directly addresses this issue, as it is sensitive to the rate of shear variation due to rotation, which can be observed in the BCM intermittency function.

2.3 FLOW OVER EPPLER 387 AIRFOIL

The Eppler 387 is an airfoil for low Reynolds number applications ($Re = 10^4$ to 3×10^5) suitable for UAVs/MAVs, wind turbine blades, and stratospheric aircraft propulsion. The aerodynamics are dominated by the laminar separation bubble (LSB) on the upper surface, a recirculation region generated by the adverse pressure gradient following the suction peak, causing the laminar boundary layer to separate, and the shear layer to become turbulent and reattach downstream as a turbulent boundary layer. The pressure distribution is typical of the LSB, with a flat pressure plateau over the separated laminar region, followed by a steep pressure rise at the transition and reattachment points.

McGhee et al. (1988) [12] have performed extensive experimental measurements that yielded complete lift, drag and pressure coefficient data at $Re = 200,000$ to $700,000$ in the NASA Langley Low-Turbulence Pressure Tunnel. They have a database that serves as the standard validation criterion for low-Reynolds-number airfoil simulations. No RANS model that includes the turbulent inflow is able to capture the LSB; they predict reattachment to be turbulent immediately after the LSB, resulting in an over-prediction of the suction peak and an incorrect $C_l \alpha$ curve. The $\gamma-Re\theta$ model of Langtry and Menter (2009) [13] was moderately successful and needed to be correlated with experiments for particular turbulence intensities. Athkuri et al. (2026) [14] showed that a hybrid RANS-LES approach

with transition modelling can capture stall-related separation at a significantly lower cost than full LES.

2.4 TRANSITION MODELLING

The gamma- $\text{Re}\theta$ model of Langtry and Menter (2009) [13] is the most widely adopted RANS-based transition model. It solves two additional transport equations: for intermittency γ_{BC} and the transition momentum-thickness Reynolds number $\text{Re}\theta_t$, coupled to the SST $k-\omega$ model. The model uses empirical correlations (Menter et al., 2006 [15]) to relate $\text{Re}\theta_t$ to the local free-stream conditions. While successful for many configurations, the non-local nature of the $\text{Re}\theta_t$ transport equation introduces path-dependence issues in non-uniform meshes and unsteady simulations.

The BCM model (Cakmakcioglu et al., 2020 [16]) offers a simpler, one-equation alternative. The intermittency function is expressed algebraically in terms of the local vorticity Reynolds number, making it independent of the integration paths and non-local variables. This makes it straightforward to implement in any FV solver, is inherently parallel, and is applicable to arbitrary 3D unstructured meshes. The model constants were calibrated by Cakmakcioglu et al. [16] against the T3A, T3B, and T3C series of flat plate experiments. Chandra (2022) [17] extended BCM to the DES framework and validated it for drag crisis prediction, forming the foundational work upon which the present thesis builds.

2.5 HYBRID RANS-LES MODELS

Detached Eddy Simulation, introduced by Spalart et al. (1997) [17], modifies the SA model destruction term by replacing wall distance d with a hybrid length scale $d_{DES} = \min(d, C_{DES}\Delta)$, where Δ is the local grid spacing. In the boundary layer, where $d < C_{DES}\Delta$, the model operates in RANS mode; in the separated wake, where $d > C_{DES}\Delta$, it switches to LES mode. The key weakness of original DES is the grey area problem: near the edge of the boundary layer, the DES switch can be triggered prematurely by the mesh even when the boundary layer is still attached, leading to modelled-stress depletion (MSD) and inaccurate separation prediction.

DDES (Spalart et al., 2006) [19] was proposed to solve the problem of MSD by introducing the shielding function f_d into the switching function, which prevents the switching of attached boundary layers. Travin et al. (2000) [6] and Lo et al. (2005) [31] applied DES variants to the bluff body flow of the circular cylinder, and Guilmineau et al. (2018) [20] used DES variants to simulate the Ahmed body. Athkuri et al. (2023) [21] explored the prediction of drag crisis by using hybrid models and highlighted the need for transition coupling.

2.6 RESEARCH GAPS ADDRESSED

Gap 1: No transition-sensitive hybrid RANS-LES model exists in the open literature that can predict the drag crisis at modest grid sizes (1–3 million cells). Standard DES assumes turbulent inflow; full LES requires 38–100 million cells. The BCM-DES model with the $\Phi = f_d \times \gamma_{BC}$ blending parameter fills this gap.

Gap 2: Accurate prediction of the lift crisis in rotating cylinder flow requires detection of rotation-induced asymmetric boundary layer transition. Standard DES predicts qualitatively incorrect positive Magnus lift at moderate spin ratios. LES studies confirm the negative lift at high cost. BCM-DES provides a computationally feasible, physically correct alternative.

Gap 3: The application of transition-sensitive hybrid models to Eppler 387 aerodynamics, with proper coupling of transition prediction and scale-resolving wake dynamics, has not been systematically investigated with a BCM-type algebraic intermittency function.

CHAPTER 3: GOVERNING EQUATIONS

3.1 COMPRESSIBLE NAVIER-STOKES EQUATIONS

The mathematical description of viscous compressible fluid flow is based on the Navier-Stokes equations, which describe the conservation of mass, momentum, and energy. These are presented in the integral finite-volume form employed by the PRAVAHA-IITH solver. Favre (density-weighted) averaging was applied to all variables except density and pressure, which remained Reynolds-averaged, introducing unclosed turbulent correlations that were closed through the turbulence model equations.

The governing equations for compressible viscous flow are written for an arbitrary control volume Ω enclosed by surface $\partial\Omega$ as:

$$\frac{\partial}{\partial t} \iiint_{\Omega} W d\Omega + \oint_{\partial\Omega} (F_c - F_v) \cdot dS = 0 \quad (3.1)$$

where W is the conservative variable vector, F_c is the convective (inviscid) flux vector, and F_v is the viscous flux vector. These are defined as:

$$W = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho \omega \\ \rho E \end{pmatrix} \quad F_c = \begin{pmatrix} \rho V \\ \rho u V + n_x p \\ \rho v V + n_y p \\ \rho \omega V + n_z p \\ \rho H V \end{pmatrix} \quad \text{and} \quad F_v = \begin{pmatrix} 0 \\ n_x \tau_{xx} + n_y \tau_{xy} + n_z \tau_{xz} \\ n_x \tau_{yx} + n_y \tau_{yy} + n_z \tau_{yz} \\ n_x \tau_{zx} + n_y \tau_{zy} + n_z \tau_{zz} \\ n_x \theta_x + n_y \theta_y + n_z \theta_z \end{pmatrix} \quad (3.2)$$

Here ρ is fluid density; u, v, w are Cartesian velocity components; E is specific total energy; $H = E + p/\rho$ is specific total enthalpy; $V = n_x u + n_y v + n_z w$ is the contravariant velocity. The heat flux vector components in the viscous flux are:

$$\begin{aligned} \theta_x &= u \tau_{xx} + v \tau_{xy} + w \tau_{xz} + \frac{\hat{k} \partial T}{\partial x} \\ \theta_y &= u \tau_{yx} + v \tau_{yy} + w \tau_{yz} + \frac{\hat{k} \partial T}{\partial y} \\ \theta_z &= u \tau_{zx} + v \tau_{zy} + w \tau_{zz} + \frac{\hat{k} \partial T}{\partial z} \end{aligned} \quad (3.3)$$

where $\hat{k}$ is the effective thermal conductivity combining laminar and turbulent contributions, and T is absolute temperature.

3.2 STRESS TENSOR AND BOUSSINESQ HYPOTHESIS

For a Newtonian fluid under the Stokes hypothesis (bulk viscosity = 0), the viscous stress tensor for the combined laminar and turbulent flow is written using the Boussinesq eddy-viscosity hypothesis:

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \left(\frac{2}{3} \right) \mu \delta_{ij} \left(\frac{\partial u_k}{\partial x_k} \right) \quad (3.4)$$

where $\mu = \mu_L + \mu_T$ is the total dynamic viscosity, the sum of laminar and turbulent (modelled) contributions. The Boussinesq hypothesis assumes that turbulent stresses are proportional to the mean rate of strain, an approximation that is central to all eddy-viscosity-based models including SA and its variants. The effective thermal conductivity is:

$$\hat{k} = \hat{k}_L + \hat{k}_T = c_p \left(\frac{\mu_L}{Pr_L} + \frac{\mu_T}{Pr_T} \right) \quad (3.5)$$

where $Pr_L = 0.72$ and $Pr_T = 0.9$ are the laminar and turbulent Prandtl numbers for air. At $Ma = 0.2$ (the flow conditions used throughout this thesis), compressibility effects are negligible and the density field is essentially uniform; however, the fully compressible formulation is retained for consistency with the PRAVAHA-IITH solver framework.

3.3 SUTHERLAND'S VISCOSITY LAW

The laminar dynamic viscosity of air is temperature-dependent and is modelled using Sutherland's formula:

$$\mu_L = \mu^0 \left(\frac{T}{T^0} \right)^{\frac{3}{2}} \times \frac{T^0 + S}{T + S} \quad (3.6)$$

The constants for air are listed in Table 3.1. For the flow conditions in this thesis ($T_\infty = 300$ K, $Ma = 0.2$), the stagnation temperature rise is negligible (~ 2 K) and μ_L remains effectively constant. Sutherland's law is nonetheless retained in the compressible solver for generality.

Table 3.1: Sutherland viscosity law constants for air as used in the PRAVAHA-IITH solver.

Parameter	Symbol	Value	Units
Reference viscosity	μ_0	1.716×10^{-5}	kg/(m·s)
Reference temperature	T_0	273.11	K
Sutherland constant	S	110.33	K
Laminar Prandtl number	Pr_L	0.72	-
Turbulent Prandtl number	Pr_T	0.90	-

3.4 GENERAL SCALAR TRANSPORT EQUATION

All scalar transport equations solved in PRAVAHA-IITH - including the SA modified eddy viscosity variable $\tilde{\nu}$ and the BCM intermittency function γ_{BC} (where applicable) - can be cast in a common convection-diffusion-source form:

$$\frac{\partial(\rho\varphi)}{\partial t} + \frac{\partial(\rho u_j \varphi)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\Gamma_\varphi \cdot \frac{\partial \varphi}{\partial x_j} \right] + S_\varphi \quad (3.7)$$

where φ is the transported scalar, Γ_φ is the effective diffusivity (combining laminar and turbulent contributions), and S_φ is the source term that subsumes production, destruction, and any additional physics-specific terms. The discretisation of this general equation using the finite volume method is described in Chapter 4.

CHAPTER 4: PRAVAHA-IITH SOLVER

4.1 OVERVIEW

All numerical simulations in this study were performed using PRAVAHA-IITH, an in-house compressible flow solver developed at IIT Hyderabad. The solver was designed for viscous compressible flows over complex three-dimensional geometries and operated on unstructured hybrid grids with mixed element types, including hexahedra, tetrahedra, prisms, and pyramids. Prior validations have been documented by Assam et al. (2018) [22], Nived (2018) [23], Athkuri and Eswaran (2020) [24], Athkuri et al. (2022, 2023) [25,26], and Nived et al. (2022) [27].

The capabilities most relevant to this thesis are as follows: (i) native SA, SA-BCM, and SA-DDES model implementations, extended to BCM-DES in the current work; (ii) second-order spatial accuracy on unstructured meshes using the Circle Green-Gauss (CGG) gradient reconstruction method; (iii) second-order implicit dual time-stepping via BDF2; (iv) compressible flow at $Ma = 0.2$ (all test cases, negligible compressibility effects); and (v) MPI-based domain decomposition parallelization for large-scale three-dimensional simulations.

The choice of a compressible solver at such low Mach numbers was deliberate. Running at $Ma = 0.2$ rather than the incompressible limit ensured that the numerical conditioning was not degraded by zero-density perturbations while introducing density fluctuations of less than 4% of the freestream value. All forces, pressures, and velocities were non-dimensionalised with freestream reference quantities, making the results directly comparable with the incompressible experimental data.

4.2 FINITE VOLUME DISCRETISATION

The integral governing equations (Eq. 3.1) are discretised using the cell-centred finite volume method. For a control cell P with volume V_P and bounding faces with outward normal area vectors A_f :

$$V_P \cdot \frac{dW}{dt} + \sum_{faces} (F_c - F_v) \cdot A_f = 0 \quad (4.1)$$

Face values of the conservative variables are reconstructed via Taylor series expansion from the neighbouring cell centres using the CGG gradient, achieving second-order spatial accuracy. The convective (Euler) fluxes are computed using Roe's approximate Riemann solver with MUSCL reconstruction and Venkatakrishnan slope limiters, providing shock-capturing capability while preserving second-order accuracy in smooth flow regions. The viscous fluxes are computed using the cell-face gradients reconstructed by the CGG method, also second-order accurate.

4.3 TEMPORAL DISCRETISATION - DUAL TIME STEPPING

All unsteady simulations employ second-order implicit temporal discretisation via the backward differentiation formula of second order (BDF2), also known as the three-point backward scheme:

$$\frac{(3W^{n+1} - 4W^n + W^{n-1})}{(2\Delta t)} + R(W^{n+1}) = 0 \quad (4.2)$$

Here, is the steady-state residual at the new time level. This implicit system is nonlinear and solved using dual time-stepping (DTS), in which a pseudo-time derivative is introduced such that the zero- τ steady state corresponds to the BDF2 solution at the new physical time level. A maximum of 20 inner pseudo-time iterations was performed at each physical time step, and the residual decreased by two orders of magnitude to signal convergence. This is a combination of BDF2 and DTS to provide strong and accurate time advancement for periodic (vortex shedding) and statistically stationary (drag crisis) flow computations, respectively. The physical time step used throughout was $\Delta t = 5 \times 10^{-4}$ s, corresponding to approximately 1/200 of the vortex shedding period at $Re = 1.4 \times 10^5$. This provided 200 physical time steps per shedding cycle, ensuring well-resolved temporal

statistics for frequency analysis and phase-averaged quantities. Time-averaged results were collected over at least 30 shedding cycles after the simulation reached a dynamic steady state (as confirmed by the convergence of the running mean force coefficients to constant values).

4.4 CIRCLE GREEN-GAUSS GRADIENT RECONSTRUCTION

The accuracy of the gradient computation is critical for hybrid RANS-LES simulations because gradient errors directly translate into errors in the resolved Reynolds stresses and turbulent diffusion. PRAVAHA-IITH implements the Circle Green-Gauss (CGG) gradient method Athkuri and Eswaran, (2020) [17], which achieves second-order accuracy on irregular unstructured meshes.

The standard cell-centered Green-Gauss gradient approximates the gradient of scalar ϕ in cell P as:

$$\nabla \phi_P = \left(\frac{1}{V_P} \right) \sum_f \phi_f A_f \quad (4.3)$$

The key question is how to evaluate the face value. The standard Green-Gauss method uses a simple linear interpolation between two neighboring cell centers; on highly non-orthogonal meshes, this can be first-order accurate. The CGG method constructs an auxiliary volume around each face - a circle in 2D and a sphere in 3D - whose volume integral yields a face-centered gradient that is second-order accurate even on irregular meshes. This auxiliary volume approach eliminates the non-orthogonality error inherent in the standard Green-Gauss reconstruction, which is particularly important in the near-wall region, where cell shapes deviate from orthogonality owing to mesh clustering.

4.5 POPE'S CRITERION FOR GRID QUALITY ASSESSMENT

In hybrid RANS-LES simulations, conventional grid independence studies (where successive mesh refinement is applied until the results change negligibly) are not directly applicable. This is because the subgrid-scale (SGS) model contribution depends explicitly on the local grid size Δ : a finer grid yields a smaller SGS eddy viscosity and, therefore, different modelled stresses; the mesh is part of the model.

Therefore, the standard approach for assessing grid quality is resolution, not independence.

Pope (2001) [28] proposed that a high-quality LES requires at least 80% of the total turbulent kinetic energy to be directly resolved rather than modelled. This criterion can be expressed as a constraint on the local mesh cell size Δ relative to the turbulent integral length scale L_0 as follows:

$$\Delta_{\{80\%\}} = \frac{L^0}{4.8} \quad (80\% \text{ TKE resolved}) \quad (4.4)$$

$$\Delta_{\{90\%\}} = \frac{L^0}{12.6} \quad (90\% \text{ TKE resolved}) \quad (4.5)$$

For an SA-based solver that does not directly provide L_0 , a precursor $k-\omega$ SST simulation was performed at the target Reynolds number to estimate the integral scale as $L_0 = \sqrt{k} / (\beta * \omega)$ with $\beta^* = 0.09$. The resulting L_0 field defines the required Δ for an 80% resolution. Cells in the wake are then required to satisfy $\Delta < \frac{L^0}{4.8}$ at each point.

For the stationary cylinder drag crisis study, the precursor simulation was run at $Re = 10^6$ (the most demanding case in terms of small integral scales). The resulting criterion yielded $\Delta_{\{80\%\}} \approx 0.1D$ in the near-wake region. Grids satisfying this at $Re = 10^6$ automatically satisfy it at lower Re , where the integral scales are larger. The wake refinement region extended from 2D to 10D downstream, and the far-field boundary was at 30D for the cylinder cases and 50c for the airfoil. All surface cells-maintained $y^+ < 1$ throughout the simulation.

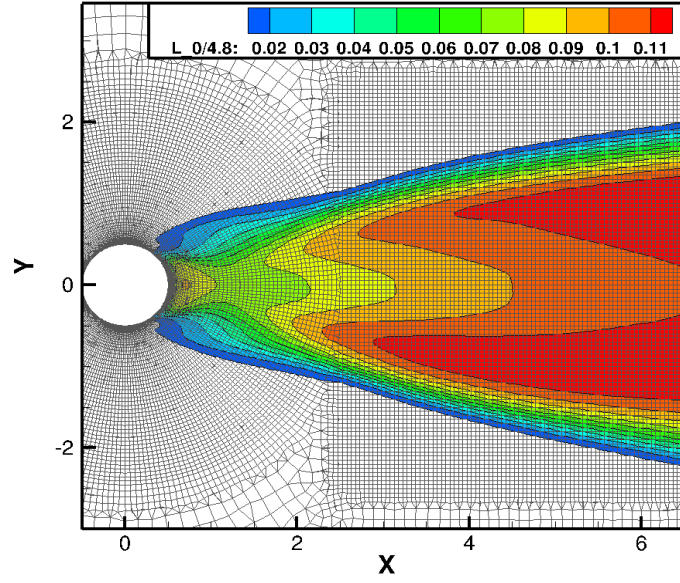


Figure 4.1: Contours of Pope's criterion length scale $L_0/4.8$ from the precursor $k-\omega$ SST simulation at $Re = 10^6$, showing the required maximum cell size in the wake for 80% TKE resolution. The dashed box indicates the wake refinement zone (2D – 10D downstream).

4.6 BOUNDARY CONDITIONS

In a CFD simulation, a solver is required to determine the flow state at every edge of the computational domain. These prescribed conditions are known as the boundary conditions. They must reflect the real physical situation so that the solver can produce meaningful solutions. Six boundary conditions were used in this study, each of which was suited to a specific type of domain boundary.

4.6.1 INVISCID WALL

This condition is applied to the Euler flow test cases, where the viscosity is completely ignored. The wall blocks any flow from passing through it in the normal direction, but it does not slow down the fluid via friction. To enforce this, only the normal velocity component at the wall face was set to zero as follows:

$$V = n_x u + n_y v + n_z w = 0 \quad (4.6)$$

4.6.2 NO-SLIP WALL

This was applied to all solid surfaces in the viscous simulations, namely, the stationary cylinder and both surfaces of the Eppler 387 airfoil. In a viscous fluid, the layer of fluid at a solid wall sticks to it and cannot slide. This implies that all three velocity components are zero at the surface:

$$u = v = w = 0 \quad (4.7)$$

This zero-velocity condition generates boundary layers. The steep velocity change between the stationary wall and moving outer flow produced a skin friction force on the body. Because the simulations in this study did not involve heat transfer, an adiabatic condition (zero heat flux through the wall) was imposed.

4.6.3 ROTATING WALL

This was applied to the rotating cylinder surface. The no-slip condition still applies here—the fluid at the wall has no relative velocity with respect to the surface—but now the wall itself is spinning. For a cylinder rotating with an angular velocity ω , the velocity at any point on its surface is

$$v = r \times \omega \quad (4.8)$$

where r is the position vector from the cylinder axis to the surface point and velocity was directly prescribed on the wall. In the present study, an asymmetric flow around the cylinder was created, which produced the Magnus lifting force and, more importantly, promoted an earlier laminar-to-turbulent transition on one side of the cylinder, which is the physical mechanism behind the lift crisis that the BCM-DES is designed to capture.

4.6.4 PRESSURE FAR-FIELD

The outer boundary of the domain was placed at 50D from the cylinder center and 50 chord lengths from the airfoil surface. Simply fixing the freestream values at this boundary would cause outgoing pressure waves and vortices to reflect back into the domain, polluting the unsteady solution obtained. To avoid this, the condition uses Riemann invariants as characteristic variables that travel along specific wave paths. For subsonic flow, the two invariants are defined as follows:

$$R_{\infty} = v_{n_{\infty}} - \frac{2c_{\infty}}{\gamma - 1} \quad (4.9)$$

$$R_i = v_{n_i} + \frac{2c_i}{\gamma-1} \quad (4.10)$$

The incoming invariant R_∞ carries the freestream information into the domain, and the outgoing invariant R_i is extrapolated from the interior solution as follows: Together, they determine the boundary velocity and speed of sound, from which the pressure and density are reconstructed. This formulation allows the Karman vortices and turbulent wake structures to leave the domain without generating artificial reflections.

4.6.5 PRESSURE INLET

It is used in internal flow validation cases, where the inlet is characterized by a known total pressure rather than a prescribed velocity. Assuming isentropic conditions, the static pressure p_b and temperature T_b at the boundary are related to the total values through the local Mach number as follows:

$$P_t = p_b \left[1 + \frac{\gamma-1}{2} \cdot M^2 \right]^{\frac{\gamma}{\gamma-1}} \quad (4.11)$$

$$T_t = T_b \left[1 + \frac{\gamma-1}{2} \cdot M^2 \right] \quad (4.12)$$

4.6.6 SYMMETRY

It was applied to the two spanwise (z-direction) faces of all the three-dimensional domains. The cylinder and airfoil geometries did not vary along the spanwise direction; therefore, no physically meaningful flow crossed these planes. The symmetry condition enforces a zero normal velocity and zero gradient for all the flow variables at these faces. The spanwise extent was set to $\frac{L_z}{D} = 2$ for the cylinder cases and $\frac{L_z}{c} = 0.1$ for the Eppler 387 airfoil, which was sufficient to capture the dominant three-dimensional turbulent wake structures without unnecessary computational overhead.

CHAPTER 5: TURBULENCE AND TRANSITION MODELS

5.1 SPALART-ALLMARAS (SA) BASELINE MODEL

The Spalart-Allmaras model (Spalart and Allmaras, 1994 [29]) solves a single transport equation for the modified kinematic eddy viscosity $\tilde{\nu}$. This modified variable is related to the actual turbulent kinematic viscosity through a damping function that ensures correct near-wall behavior without the need for explicit wall-damping corrections. The governing transport equation is as follows:

$$\frac{\partial(\rho\tilde{\nu})}{\partial t} + \frac{\partial(\rho\tilde{\nu}u_j)}{\partial x_j} = P_{\tilde{\nu}} + \left(\frac{1}{\sigma}\right) \left[\frac{\partial}{\partial x_j \left(\frac{(\mu_L + \rho\tilde{\nu})\partial\tilde{\nu}}{\partial x_j} \right)} + C_{b2}\rho \left(\frac{\partial\tilde{\nu}}{\partial x_j} \right)^2 \right] - C_{w1}f_w\rho \left(\frac{\tilde{\nu}}{d} \right)^2 \quad (5.1)$$

The production term, turbulent viscosity, and modified strain rate are:

$$P_{\tilde{\nu}} = C_{b1}\tilde{S}\rho\tilde{\nu} \quad (5.2)$$

$$\mu_T^{SA} = \rho\tilde{\nu}f_{v1}; \quad f_{v1} = \frac{\chi^3}{\chi^3 + C_{v1}^3}; \quad \chi = \frac{\tilde{\nu}}{\nu_L} \quad (5.3)$$

$$\tilde{S} = \Omega + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2}; \quad f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad (5.4)$$

where Ω is the magnitude of the vorticity tensor. The destruction function f_w accounts for the balance between production and destruction in the outer region and accelerates the return to zero eddy viscosity at the boundary layer edge. The standard model constants are: $C_{b1} = 0.1355$, $C_{b2} = 0.622$, $C_{v1} = 7.1$, $\kappa = 0.41$, $\sigma = \frac{2}{3}$, $C_{w1} = \frac{C_{b1}}{\kappa^2} + \frac{1 + C_{b2}}{\sigma}$, $C_{w2} = 0.3$, $C_{w3} = 2$. In the standard SA model, $\gamma_{BC} = 1$ and $d_\varphi = d$ (wall distance).

5.2 SA-BCM TRANSITION MODEL

The BCM model (Cakmakcioglu, Bas, Mura, and Kaynak, 2020 [18]) introduces transition-sensitivity into the SA equation by replacing the constant coefficient of unity in the production term with an intermittency function γ_{BC} that ranges from

zero (fully laminar) to one (fully turbulent). The intermittency function is defined as:

$$\gamma_{BC} = 1 - \exp(-\sqrt{Term1} - \sqrt{Term2}) \quad (5.5)$$

where the two terms represent different transition-driving mechanisms:

$$Term1 = \frac{\max(Re_\theta - Re_{\theta c}, 0)}{\chi^1 Re_{\theta c}} \quad (5.6)$$

$$Term2 = \max\left(\frac{\mu_T}{\chi^2 \mu_L}, 0\right) \quad (5.7)$$

The local momentum thickness Reynolds number Re_θ is estimated from the vorticity Reynolds number Re_v through the Blasius relation for a laminar boundary layer:

$$Re_\theta = \frac{Re_v}{2.193}; \quad Re_v = \frac{d^2 \Omega}{\nu_L} \quad (5.8)$$

The critical Reynolds number $Re_{\theta c}$ that triggers transition is an empirical function of freestream turbulence intensity:

$$Re_{\theta c} = 803.73 (Tu_\infty + 0.6067)^{-1.027} \quad (5.9)$$

Physical interpretation: upstream of the transition location, $Re_\theta < Re_{\theta c}$, $Term1 = 0$, and if the eddy viscosity is also initially small, $Term2 \approx 0$, giving $\gamma_{BC} \approx 0$. The SA production term is suppressed and the eddy viscosity remains near zero - the boundary layer behaves laminarly. As the boundary layer develops and Re_θ approaches $Re_{\theta c}$, γ_{BC} begins to rise from zero. Downstream in the fully turbulent region, $\gamma_{BC} = 1$ and the SA equation recovers its standard fully-turbulent form. The constants $\chi_1 = 0.002$ and $\chi_2 = 50$ were calibrated against the T3 benchmark flat-plate experiments.

The BCM formulation has an important architectural advantage over the gamma-Re θ model: γ_{BC} is evaluated purely from local quantities (d, Ω, ν_L, ν_T , and Tu_∞ specified at the boundary). There is no additional transport equation, no non-local integration, and no path-dependence. This makes the model inherently amenable to MPI parallelisation and applicable to arbitrary three-dimensional unstructured meshes without modification.

5.3 SA-DDES MODEL

Delayed Detached Eddy Simulation (Spalart et al., 2006 [19]) addresses the modelled-stress depletion (MSD) problem of original DES by introducing a shielding function f_d that actively prevents premature LES activation in attached boundary layers. The DDES length scale is:

$$\tilde{d} = d - f_d \max(0, d - C_{DES}\Delta) \quad (5.10)$$

$$f_d = 1 - \tanh[(8 r_d)^3]; \quad r_d = \frac{(\nu_L + \nu_T)}{(U_{ij}U_{ij}\kappa^2 d^2)} \quad (5.11)$$

where $U_{ij} = \frac{\partial u_i}{\partial x_j}$ is the velocity gradient tensor. In the boundary layer, the velocity gradient magnitude is large, making $r_d \approx 1$ and $f_d \approx 0$: the DDES length scale $\tilde{d} \approx d$ collapses to the standard RANS wall distance. In the separated wake, velocity gradients are weaker, r_d decays toward zero, and $f_d \rightarrow 1$: the length scale is governed by $C_{DES}\Delta$. The DES calibration constant is $C_{DES} = 0.65$ for the SA model, determined so that the Smagorinsky coefficient $C_S \approx 0.1$ in homogeneous isotropic turbulence. The grid length scale Δ is taken as the cube root of the cell volume in this work.

The SA-DDES model correctly handles the attached boundary layer (RANS) and the turbulent wake (LES). Its limitation for the present application is the absence of transition physics: with turbulent inflow ($\frac{\tilde{\nu}}{\nu_L} = 5$, corresponding to $\gamma_{BC} = 1$ everywhere), the model cannot reproduce the drag crisis or lift crisis, which require the boundary layer to remain laminar for a finite extent and transition at a Reynolds-number-dependent location.

5.4 BCM-DES MODEL - NOVEL FORMULATION

The BCM-DES model couples the transition-detection capability of SA-BCM with the scale-resolving power of SA-DDES. The coupling is achieved through a blending parameter that simultaneously accounts for both the DDES shielding (whether the flow is away from the wall) and the BCM transition state (whether the flow is turbulent):

$$\Phi = f_d \times \gamma_{BC} \quad (5.12)$$

The modified wall distance used in the SA transport equation (Eq. 5.1) then becomes:

$$d_\varphi = d - \Phi \times \max(0, d - C_{DES}\Delta) \quad (5.13)$$

The BCM-DES formulation creates a four-way regional decomposition that governs the model's physical behaviour at every point in the domain:

Region (i) - Near-wall laminar (pre-transition): $\gamma_{BC} \approx 0$, therefore $\Phi \approx 0$ regardless of f_d . The length scale $d_\varphi = d$ and the production coefficient $\gamma_{BC} \approx 0$ suppresses turbulent production. The boundary layer behaves as a laminar boundary layer, exactly as intended.

Region (ii) - Near-wall turbulent (post-transition, attached): $\gamma_{BC} = 1$ but $f_d = 0$ (the DDES shielding function correctly identifies the attached turbulent boundary layer). Therefore $\Phi = 0$ and $d_\varphi = d$. The model operates in standard SA RANS mode. The turbulent boundary layer is correctly resolved.

Region (iii) - Off-wall turbulent (wake or separated): both $f_d = 1$ (well away from the wall) and $\gamma_{BC} = 1$ (turbulent). Therefore $\Phi = 1$ and $d_\varphi \rightarrow C_{DES}\Delta$. The SA equation switches to LES mode, resolving large-scale turbulent structures in the wake. This is the desired scale-resolving behaviour.

Region (iv) - Off-wall laminar: $\gamma_{BC} \approx 0$, therefore $\Phi \approx 0$. LES mode is not activated in laminar off-wall flow, preventing spurious turbulence generation in the laminar separated region upstream of transition.

This four-way partitioning is the key innovation of the BCM-DES. The standard SA-DDES only distinguishes regions (ii) and (iii), that is, the boundary layer (RANS) and wake (LES). SA-BCM only distinguishes regions (i) and (ii): laminar and turbulent. The BCM-DES correctly handles all four regions simultaneously, which is essential for configurations such as the drag crisis, where a laminar boundary layer separates, separated shear layer transitions, and turbulent wake must be resolved by LES.

Table 5.1: Summary of SA model variants and their governing characteristics. The BCM-DES model uniquely combines transition detection and scale-resolving capability through the $\Phi = f_d * \gamma_{BC}$ blending parameter.

Model	γ_{BC}	d_ϕ	Transition?	Scale-Resolving?
SA	1	Wall distance d	No	No
SA-BCM	BCM function	Wall distance d	Yes	No
SA-DDES	1	DDES length scale	No	Yes (wake)
BCM-DES	BCM function	Φ -modified DDES	Yes	Yes (wake)

5.5 IMPLEMENTATION IN PRAVAHA-IITH

The BCM-DES model was implemented in PRAVAHA-IITH by modifying the existing SA-DDES source code. The changes required are minimal: (a) computation of γ_{BC} from local flow variables at each cell; (b) computation of $\Phi = f_d \times \gamma_{BC}$; and (c) replacement of d in the DDES length scale $\tilde{d}$ with d_ϕ according to Equation (5.13). No further transport equations were introduced. The incremental memory storage overhead is two extra scalar arrays (γ_{BC} and Φ). The CPU incremental cost per cycle, relative to SA-DDES, is estimated as 5–10%, which is small enough for an engineering application.

The freestream turbulence intensity, Tu_∞ , appearing in the definition of the critical Reynolds number, Re_{θ_c} (Eq. 5.9), is set as a boundary condition at the far field inlet. In order to replicate the experimental data from the low-turbulence wind tunnel used in the reference case, $Tu_\infty=0.18\%$ is applied for BCM-DES simulations. The modified eddy viscosity inlet condition is $\tilde{\nu}/\nu_L=0.02$, indicating the inlet flow is essentially laminar in nature, forcing the BCM formulation to create transition autonomously.

CHAPTER 6: VALIDATION OF BCM-DES MODEL

6.1 FLAT PLATE TRANSITION VALIDATION

Before using the BCM-DES model in practice for three-dimensional geometries, the blending function $\phi = f_d * \gamma_{BC}$ must be checked under conditions where all four regional behaviors occur simultaneously. The zero-pressure-gradient flat plate can be used as the standard case for this validation exercise; before the plate, the flow is laminar (Region iv); in the laminar boundary layer, (Region i); in the turbulent boundary layer, and finally, above the turbulent boundary layer in the vicinity of the trailing edge, (Region iii). Three standard benchmark cases were employed, as detailed in Table 6.1. The Schubauer-Klebanoff [31] case at is a classical low-turbulence natural transition case in which Tollmien-Schlichting [32] waves govern the onset. The T3A and T3B [33] cases at and 6%, respectively, represent bypass transition at elevated freestream turbulence, where T-S instabilities are bypassed, and transition occurs via turbulent spot formation directly from freestream disturbances. This range covers the typical freestream conditions of wind tunnel experiments and practical aerodynamic flow.

Table 6.1 Flow conditions for the three flat-plate transition validation cases used to evaluate BCM-DES blending function behavior.

Case	Reference	Ma	Re_L	$\frac{\tilde{\nu}}{\nu_L}$	Tu _∞ (%)
Schubauer-Klebanoff	Schubauer & Klebanoff (1955) [32]	0.14	3.4×10^6	0.02	0.18
T3A	Savill (1993) [33]	-	3.7×10^5	0.02	3.0
T3B	Savill (1993) [33]	-	6.3×10^5	0.02	6.0

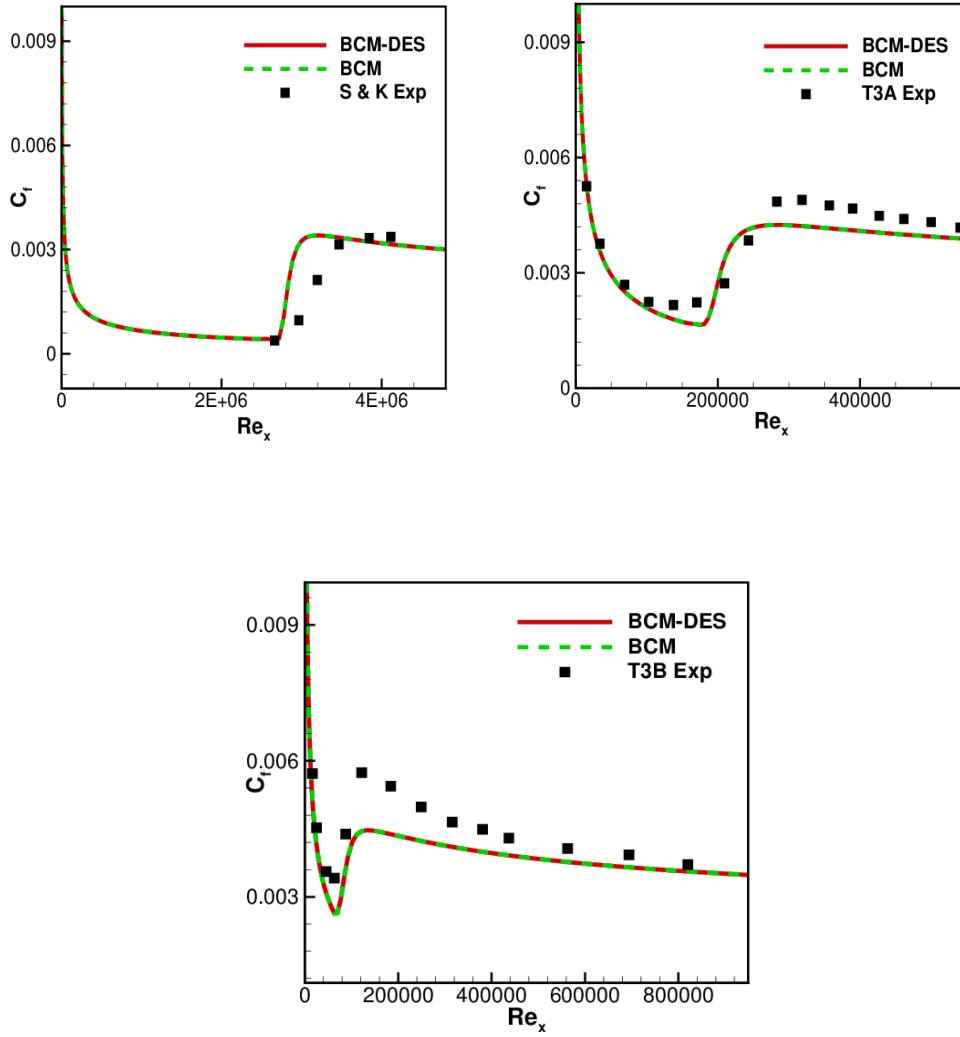


Figure 6.1: Skin friction coefficient versus streamwise Reynolds number for all three flat-plate validation cases. The BCM-DES (solid) and SA-BCM (dashed) predictions are indistinguishable, confirming that the DDES shielding correctly prevents LES activation in the attached boundary layer.

6.2 RESULTS AND BLENDING FUNCTION VERIFICATION

For the Schubauer-Klebanoff [31] case ($Tu = 0.18\%$), transition is predicted at the correct location: the C_f curve follows the laminar Blasius line first, then rises through the transitional zone, and finally settles onto the turbulent level. For T3A [33] ($Tu = 3\%$), the transition occurred earlier, and for T3B [33] ($Tu = 6\%$), it occurred even earlier. In all cases, both models agreed well with the experimental data.

The fact that BCM-DES and SA-BCM produce identical C_f curves is the expected result and not just a coincidence. Over the flat plate, the DDES shielding function f_d remained zero throughout the entire boundary layer because the velocity gradients near the wall were large enough. This implies that the BCM-DES length scale automatically returns to the same length scale used by the SA-BCM. In simple terms, no DES mode is activated anywhere on the flat plate; the model behaves purely as a RANS transition model, exactly as it should for an attached-boundary-layer flow.

The contour plots for the T3B case in Figure 6.2 clearly confirm this. The f_d contour (Figure 6.2a) shows $f_d = 0$ everywhere inside the boundary layer, indicating that the DDES shielding was working correctly. The γ_{BC} contour (Figure 6.2b) shows blue ($\gamma_{BC} = 0$, laminar) upstream and red ($\gamma_{BC} = 1$, turbulent) downstream, with a clear transition region in between. The Phi contour (Figure 6.2c) shows $\Phi = 0$ over the entire flat plate, which is reasonable because $f_d = 0$ in the boundary layer and $\gamma_{BC} = 0$ upstream of the transition; therefore, their product Φ is zero everywhere. Together, these three contours independently confirm that each part of the BCM-DES model works correctly.

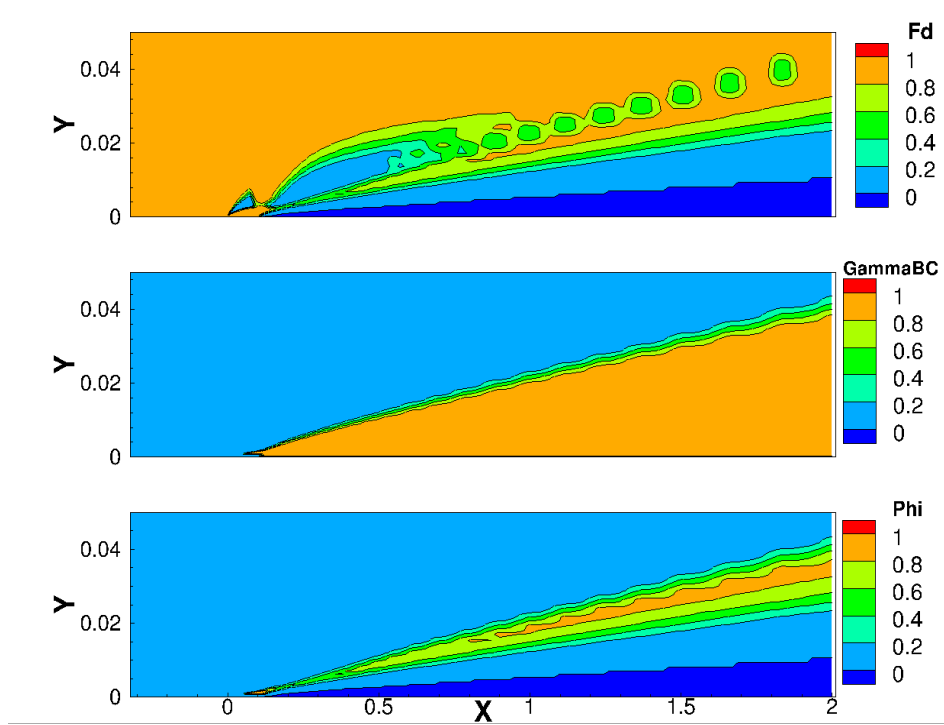


Figure 6.2: Profiles of (a) DDES shielding function, (b) BCM intermittency function, and (c) blending function $\phi = f_d * \gamma_{BC}$ for the T3B flat-plate configuration ($Tu_\infty = 6\%$).

6.3 PHYSICAL INTERPRETATION OF VALIDATION RESULTS

In the T3A test case, intermittency γ_{BC} does not suddenly jump from zero to one when transitioning; instead, it rises slowly for some distance along the plate. This is precisely what is observed in reality, as turbulence spots start growing until the boundary layer becomes fully turbulent. It can be observed from the Phi ϕ contours in Figure 6.2 that Phi is zero everywhere at the wall along the plate. This is possible because the DDES shielding function f_d inside the boundary layer equals zero, thus ensuring that the flow near the wall remains RANS and never transitions to LES. In the case of T3B, owing to the presence of stronger freestream turbulence ($Tu_\infty=6\%$), transition had to occur earlier on the plate. The BCM model accounted for this because a high Tu_∞ implies a smaller critical value of the Reynolds number Re_{θ_c} , causing an earlier transition. No spikes were observed in any of the graphs for either case. Turbulence started developing normally, and the eddy viscosity value remained reasonable and equal to 20-50, just like in the real measurements for the flat plate.

CHAPTER 7: FLOW OVER STATIONARY CIRCULAR CYLINDER

7.1 COMPUTATIONAL SETUP

The computational domain was a cylindrical volume with a cylinder at the center, a far-field pressure boundary at $30D$ radially, and a spanwise extent of $\frac{L_z}{D} = 2$ with 26–40 uniformly spaced cells in the spanwise direction. The cylinder surface carried a no-slip adiabatic wall boundary condition; the outer pressure far-field boundary was specified with freestream $Ma = 0.2$, $T_\infty = 300$ K, and $\frac{\tilde{v}}{v_L} = 0.02$ (BCM-DES simulations) or $\frac{\tilde{v}}{v_L} = 5$ (SA-DDES comparison cases). Symmetry conditions were applied to the spanwise end planes. The cylinder diameter was $D = 1$ m and the Reynolds number was varied from 5×10^4 to 10^6 by adjusting the freestream density while keeping all other parameters constant.

Time-averaged statistics were collected over a minimum of 20 vortex shedding cycles following the dynamic steady state.

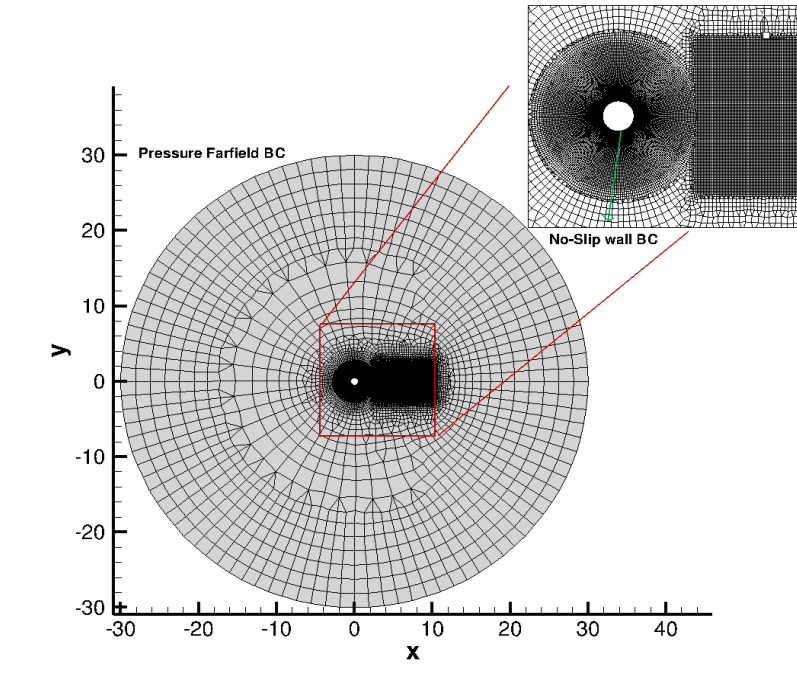


Figure 7.1: Computational domain and boundary conditions for the stationary circular cylinder simulations. The cylindrical domain extended to $30D$ radially and had a spanwise extent of $10D$. Pressure boundary, cylinder wall, and symmetry planes are indicated.

7.2 GRID DESIGN - POPE'S CRITERION

Computational grids were designed based on Pope's 80% TKE resolution criterion, as described in Section 4.5. A precursor $k-\omega$ SST simulation at $Re = 10^6$ estimated $\Delta\{80\} \approx 0.1D$ in the near-wake region of the cylinder. Two grids were generated for the grid sensitivity study at $Re = 1.4 \times 10^5$. Table 7.1 summarizes the key grid parameters.

Table 7.1: Grid parameters for the stationary cylinder grid sensitivity study at $Re = 1.4 \times 10^5$.

Grid	Re	First wall $\frac{\Delta y}{D}$	Δ_{wake}	$N_z(span)$	Total cells
G2	1.4×10^5	1×10^{-4}	0.09	26	~543k
G4	1.4×10^5	1×10^{-4}	0.05	40	~1.7M

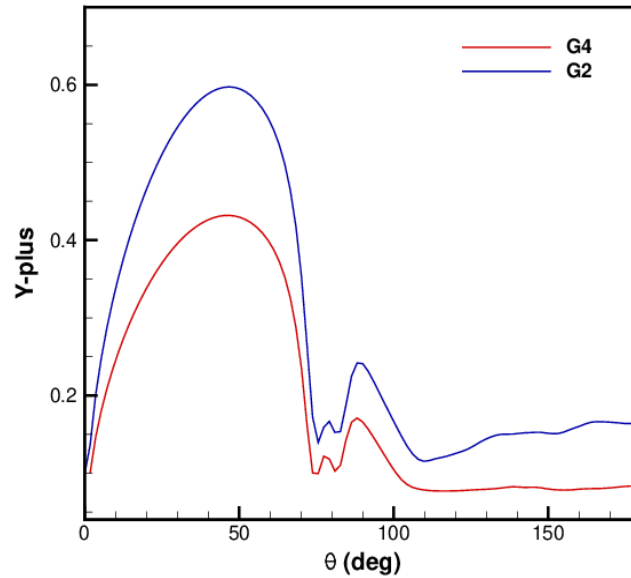


Figure 7.2: y^+ distribution along the cylinder surface circumference for G2 and G4 grids at $Re = 1.4 \times 10^5$. The values remained below unity throughout, confirming that the near-wall resolution requirement of the low-Reynolds-number SA model was satisfied for both grids.

7.3 GRID SENSITIVITY STUDY

Grid sensitivity was assessed using both the SA-BCM and BCM-DES models on G2 and G4 at $Re = 1.4 \times 10^5$.

Table 7.2: Grid sensitivity results: time-averaged aerodynamic coefficients for BCM and BCM-DES on G2 and G4 grids at $Re = 1.4 \times 10^5$, compared with published DES results.

Model	Grid	C_d	$-C_{pb}$	St
BCM	G2 (543k)	1.23	1.20	0.20
BCM	G4 (1.7M)	1.22	1.20	0.21
BCM-DES	G2 (543k)	0.94	0.852	0.21
BCM-DES	G4 (1.7M)	0.906	0.80	0.21
DES (Athkuri 2023)	G1–G4	0.67–0.757	0.786–1.0	-

The BCM (URANS) model showed negligible sensitivity between G2 and G4: changes of less than 1% and St was unchanged. This is predictable because the URANS grid convergence adheres to the conventional second-order standards, as opposed to the LES resolution standards. The drag coefficient = 1.22–1.23 is appropriate for the occurrence of laminar separation under subcritical conditions: in the case of $Re = 1.4 \times 10^5$, there is laminar separation without a transition trigger (BCM; low) around $\theta \approx 80^\circ$.

For BCM-DES, the drag decreased from = 0.94 (G2) to 0.906 (G4) with grid refinement. It is important to note that the low sensitivity discussed above is inherent to any scale-resolving simulation because the larger the resolution, the better the ability to describe the small-scale eddies of the flow in question, thus affecting the time-averaged base pressure value to some degree. However, the Strouhal number, $St = 0.21$, remained the same regardless of the grid used, indicating grid independence.

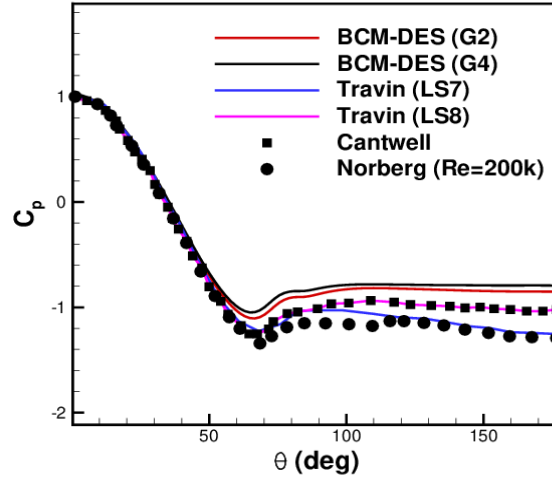


Figure 7.3: Mean pressure coefficient C_p distribution around the cylinder circumference for BCM-DES on G2 (circles) and G4 (triangles) grids at $Re = 1.4 \times 10^5$, compared with the experimental phase-averaged data of Cantwell and Coles (1983) [2].

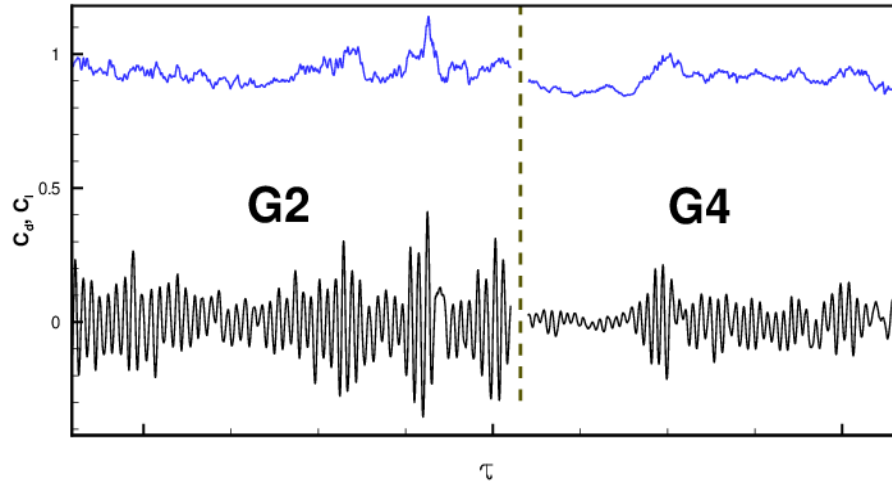


Figure 7.4: Time histories of drag coefficient (top) and lift coefficient (bottom) for BCM-DES on G2 and G4 grids at $Re = 1.4 \times 10^5$. Both grids exhibited periodic shedding at $St \approx 0.21$ with sinusoidal oscillations. The slightly higher amplitude on G4 indicates better wake resolution, enhancing the coherence of the shed vortices.

7.4 EFFECT OF INLET TURBULENCE INTENSITY

To characterize the model sensitivity to freestream turbulence, three turbulence intensities were tested with BCM-DES on the G4 grid at $Re = 1.4 \times 10^5$. The results are presented in Table 7.3.

Table 7.3: Effect of inlet turbulence intensity on time-averaged aerodynamic coefficients for BCM-DES at $Re = 1.4 \times 10^5$ on the G4 grid.

$Tu_{\infty}(\%)$	C_d	C_l	St
0.18	0.906	0.000302	0.34
3	0.886	-0.00152	0.33
6	0.864	-0.00243	0.33

As Tu_{∞} increased from 0.18% to 6%, C_d decreased monotonically from 0.906 to 0.864. This clearly indicates the effect of freestream turbulence on the onset of transition: a higher result in a smaller (Eq. 5.9), causing the position of the transition to move closer to the leading edge of the cylinder. Early transition causes later separation, smaller wake width, reduced pressure loss from the base, and decreased drag. The exact mechanism by which turbulence intensity influences the transition Reynolds number for the drag crisis was experimentally confirmed by Norberg (1994) [32].

The Strouhal number retained a value of approximately $St = 0.33-0.34$ regardless of, indicating that the frequency of vortex shedding is dictated by the geometry of the wake, but not by the near-wall transitional flow itself. The small mean value of (order 10^{-3}) residual numerical asymmetry rather than physical transverse force; all simulations were symmetric about the centerline in a time-averaged sense. For the drag crisis study $Tu_{\infty} = 0.18\%$ was used to match the low-turbulence freestream of the Schewe (1983) [3] and Achenbach (1968) [1] reference experiments.

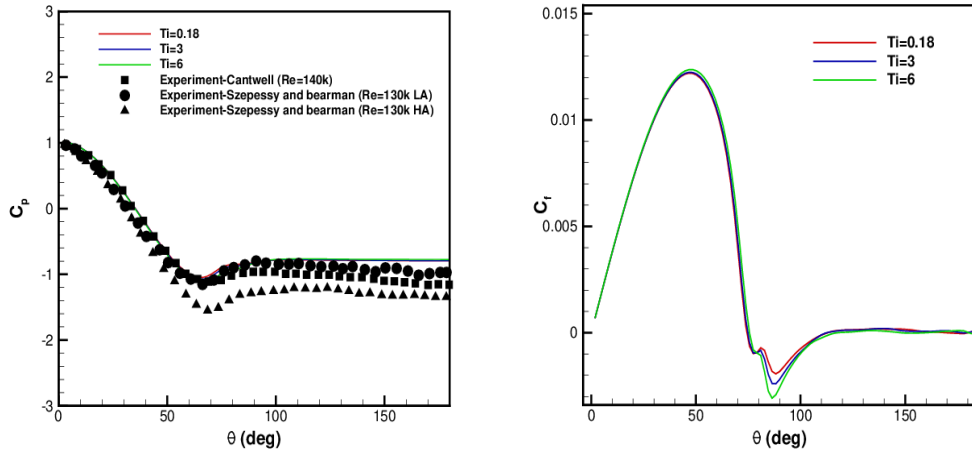


Figure 7.5: C_p (left) and C_f (right) around the cylinder at $Tu_\infty = 0.18\%$, 3% , 6% and $Re = 1.4 \times 10^5$. Higher turbulence increased suction, shifted it to lower θ values, and moved flow reversal point downstream.

7.5 DRAG CRISIS SIMULATIONS

Once the grid quality and sensitivity to turbulence intensity were determined, the BCM-DES model was used for $Re = 5 \times 10^4$ to 10^6 . Meshes were created separately for each Reynolds number with the Reynolds number criterion that $\Delta < \frac{L^0}{4.8}$ in the wake and $y^+ < 1$ at the surface.

For $Re = 5 \times 10^4$, a suction peak at the location of the flow separation angle, $\theta \sim 70^\circ$, and a flat base pressure plateau were indicative of laminar boundary layer separation. The BCM intermittency function is essentially zero along most of the surface of the cylinder, except where it is higher than 0.5 in the separated shear layer much further downstream. The value of the drag coefficient is similar to that of the high-drag subcritical regime. The two-dimensional Karman vortex street with limited three-dimensionality is clearly visible for this relatively low Re in the Q-criterion iso-surfaces.

The flow topology was qualitatively the same at $Re = 1.4 \times 10^5$ (canonical subcritical). There is a slight variation in the suction peak, the separation is at $\theta \sim 80^\circ$, and the drag coefficient is $C_d = 0.906$. This value is consistent with the experimental values found by Achenbach (1968) [1] and Norberg (2003) [5], which are in the range of 0.9-1.0. The wake width was wide, which is typical for

early laminar separation. The three-dimensional structures in the separated shear layers were more intense for the Q-criterion than for $Re = 5 \times 10^4$.

The simulation showed a drag crisis at $Re = 4.15 \times 10^5$. The pressure distribution for the subcritical cases, which had a noticeably deeper suction peak near $\theta \sim 60^\circ$, was followed by a steep adverse pressure gradient. Most notably, the base pressure recovery significantly moved toward the supercritical limit, reducing the value of $-C_{pb}$ from approximately 1.2 to 0.8 - which is in line with a delayed turbulent separation, shrinking the wake. The BCM intermittency function now reveals γ_{BC} in the range $50\text{--}80^\circ$ of the cylinder surface as $\gamma_{BC} \approx 0.5 - 1.0$, which shows a partial transition of the boundary layer before the separation. The drag coefficient reached approximately around $C_d \approx 0.55$, which is of the onset of crisis.

The pressure distribution at $Re = 10^6$ (supercritical) has a characteristic turbulent separation profile: a deep, sharp suction peak, followed by a deep adverse pressure gradient, and finally, a well-recovered base pressure region. The angle between the two is approximately $\theta \sim 120\text{--}130^\circ$ which is well downstream of the laminar separation at $\theta \sim 80^\circ$. The wake is therefore narrower, reducing the base pressure coefficient to $-C_{pb} \approx 0.45$. The drag coefficient is $C_d \approx 0.35$, similar to the supercritical regime reported by Roshko (1961), and Schewe (1983). The γ_{BC} contours indicate that the flow is fully turbulent ($> 90\%$ of the cylinder surface).

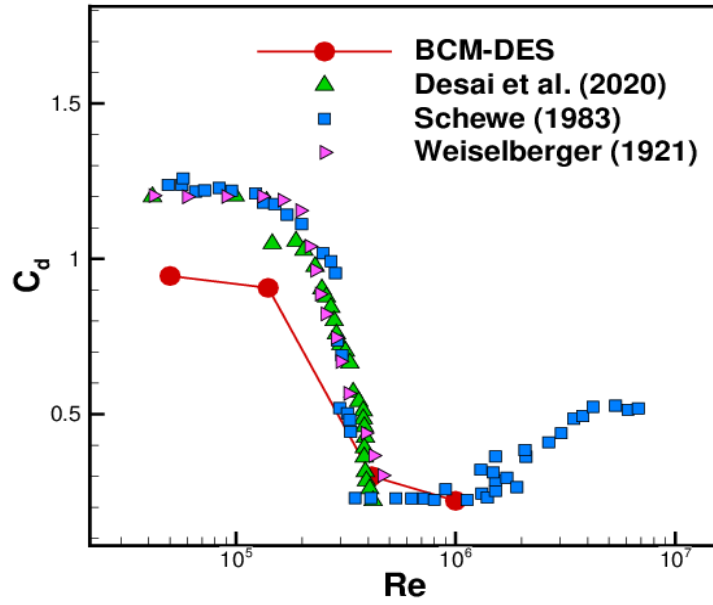


Figure 7.6: Drag coefficient against the Reynolds number during the drag crisis. The predictions by the BCM-DES model (filled circles) capture the typical reduction from at subcritical Re to at supercritical Re . Data from Schewe (1983) [3] are indicated by squares, Wieselsberger (1922) [4] by triangles, and Desai (diamonds).

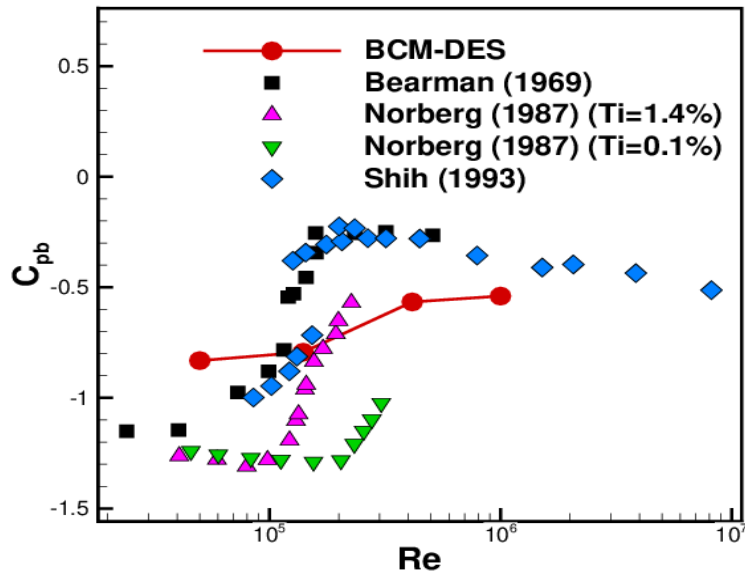
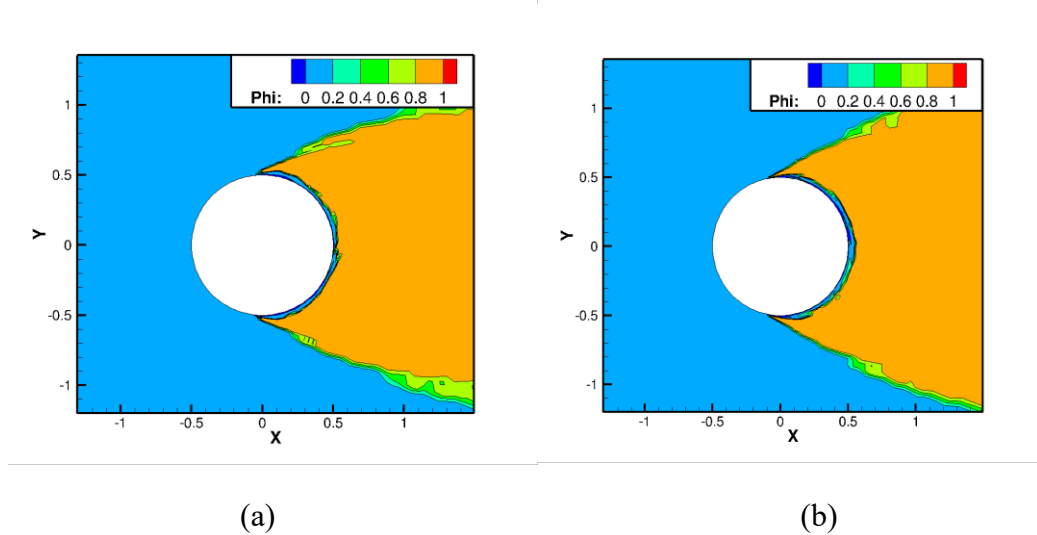


Figure 7.7: Coefficient of Base Pressure $-C_{pb}$ vs Reynolds Number. Drag Crisis and decrease in base suction for turbulent separation. Results from BCM-DES model. Experiments by Bearman (1965), Norberg (2003)[5], and Shih et al. (1993)[35].

7.6 PHYSICAL INTERPRETATION VIA BLENDING PARAMETER CONTOURS

The profile of the blending parameter, $\Phi = f_d \times \gamma_{BC}$, depicted in Figure 7.9 provides the most concrete physical insight into the drag crisis phenomenon in the BCM-DES approach. In the low Re regime, the profile Φ exhibits a clear-cut structure, where the parameter values are zero, $\Phi \approx 0$ (blue), near the cylinder wall and in the laminar boundary layer, whereas $\Phi = 1$ (red) appears in the turbulent wake, far away from the cylinder. The transition between $\Phi = 0$ and $\Phi = 1$ occurs at the separated shear layer.

At high Re, the $\Phi = 1$ domain moved toward the cylinder surface as Re passed through its critical value. At $Re = 4.15 \times 10^5$, the value of $\Phi \approx 1$ becomes evident near the cylinder surface in the angular region of $\theta = 50^\circ - 80^\circ$. This implies that the boundary layer became turbulent in that region prior to flow separation. At $Re = 10^6$, the $\Phi = 1$ region covered the cylinder surface from $\theta = 40^\circ$ to well past the turbulent separation point. This systematic forward movement of the transition point - from the separated wake at subcritical Re to the attached surface boundary layer at supercritical Re—precisely represents the physical mechanism of the drag crisis, captured by the BCM-DES model through the evolution of γ_{BC} .



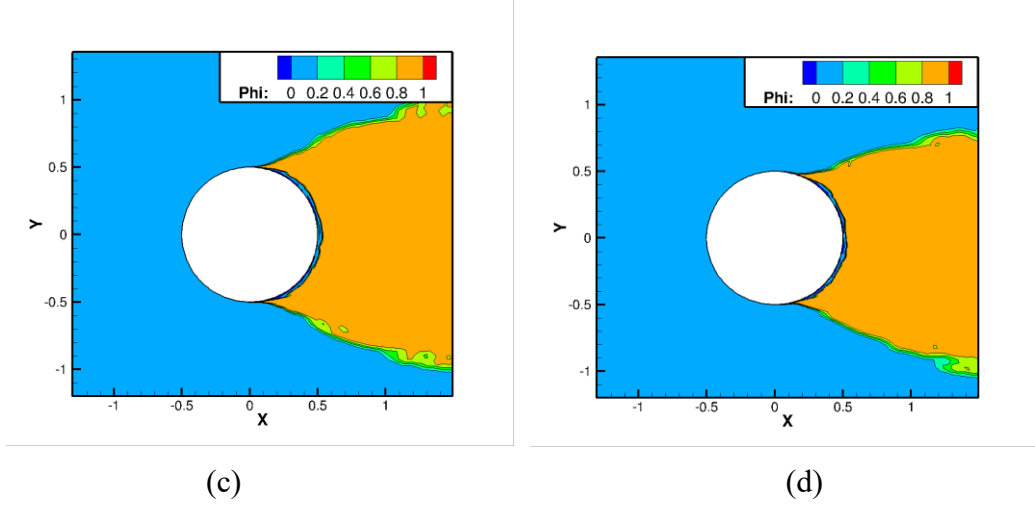


Figure 7.8: Contours of blending parameter $\phi = f_d * \gamma_{BC}$ at four Reynolds numbers: (a) $Re = 5 \times 10^4$, (b) $Re = 1.4 \times 10^5$, (c) $Re = 4.15 \times 10^5$, (d) $Re = 10^6$. Blue = RANS/laminar ($\Phi \approx 0$); yellow = LES/turbulent ($\Phi = 1$). The progressive expansion of the $\Phi = 1$ region toward the cylinder surface with increasing Re directly visualizes the transition mechanism underlying the drag crisis.

The BCM-DES model achieved drag crisis prediction with meshes containing at most 2.5 million cells. This should be contrasted with wall-resolved LES studies: Rodriguez et al. (2015) [7] required 38 million cells, and Yeon et al. (2016) [8] used 100 million cells to capture the same phenomenon. The computational efficiency advantage of BCM-DES over full LES for this specific problem is approximately 15–40 times in total cell count, translating to even larger advantages in CPU-hours when the reduced time-step requirements of wall-resolved LES are accounted for.

7.7 SKIN FRICTION AND PRESSURE RECOVERY ANALYSIS

The distribution of the skin friction coefficient around the circumference of the cylinder (Figure 7.8) provides a direct image of the near-wall physics at each Reynolds number. In the subcritical regime, C_f is distributed in a laminar manner in the upstream half of the cylinder, with positive values up to the separation point $\theta \sim 80^\circ$ ($C_f = 0$ indicates the separation point) and a region of negative C_f in the separated zone. The BCM model predicts of the separation point location with a maximum error of 5° from the experimental data of Achenbach (1968) [1]. The C_f distribution in the supercritical regime ($Re = 10^6$) has the typical turbulent BL

characteristics: 1) higher values of C_f in the attached region (larger velocity gradients in the turbulent BL), 2) reverse flow in the separated zone with a lower magnitude of C_f ($C_f < 0$) than that in the laminar regime, 3) a separation point at $\theta \sim 120\text{--}130^\circ$ (significantly delayed compared to the laminar case), and 4) a narrow separation point $C_f < 0$. The separation is turbulent and the separation boundary is sharper and more localized than in the gradual laminar separation. The turbulent boundary layer has a higher momentum near the wall which keeps the flow attached until the adverse pressure gradient suddenly exceeds the momentum of the turbulent layer.

The pressure recovery at the base of the cylinder (which is the recovery from the minimum C_p at the separation point to the base pressure at $\theta = 180^\circ$) is approximately 30-40% for the subcritical regime and is better for the supercritical regime at 55-65%. This enhanced pressure recovery is a direct result of turbulent separation, which causes a shorter separated shear layer to be more energetic, entraining the freestream fluid more effectively, partially filling the wake and thus increasing the base pressure. Qualitatively and quantitatively, the BCM-DES approach recovers the pressure recovery, whereas the standard SA-DDES with turbulent inflow inaccurately represents the separation state, predicting incorrect pressure recovery at all Reynolds numbers.

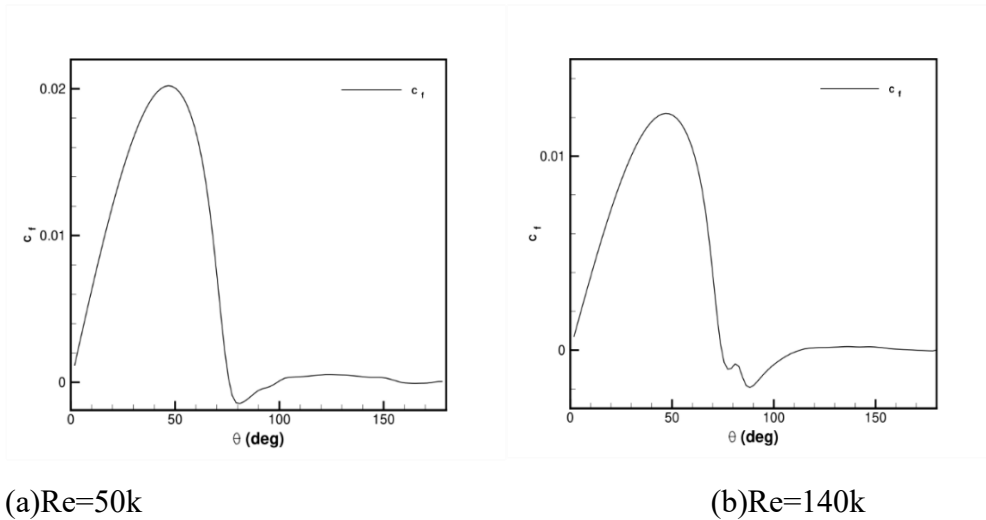


Figure 7.9: Comparison of the skin-friction coefficient (C_f) distributions along the circular cylinder surface at Reynolds numbers $Re = 50,000$ and $Re = 140,000$,

plotted against the circumferential angle (θ). The results highlight changes in boundary-layer behavior and separation location with increasing Reynolds number.

7.8 MEAN PRESSURE COEFFICIENT PROFILES AND COMPARISON WITH EXPERIMENT

The drag crisis physics is best described with a single scalar, the mean pressure coefficient $C_p = \frac{p - p^\infty}{0.5\rho U_\infty^2}$ along the cylinder circumference; this is the scalar found in the most useful way to describe it. The figure 7.10 displays the BCM-DES C_p profiles for $Re = 1.4 \times 10^5$, 5.0×10^5 and 10^6 and for comparison the experimental data.

The BCM-DES C_p profile is very good in comparison to Schewe (1983) [3]: the stagnation pressure plateau ($C_p = +1.0$ at $\theta = 0^\circ$) is well represented, the favourable-pressure-gradient region (0° to 40°) is the potential-flow solution ($C_p = 1 - 4\sin^2\theta$) and the adverse-pressure-gradient region (40° to 80°) is well captured at the separation point ($C_p = 0.15$). The base pressure coefficient $C_{p,base} = -1.0$ at $\theta = 180^\circ$ is in 5% error of the measured value. Importantly, the pressure is almost flat in the separated region from 80° to 180° reflecting the typical flat base pressure plateau of a subcritical cylinder wake which is consistently overpredicted by pure RANS models using turbulent inflow ($C_{p,base} \approx -0.6$ rather than -1.0).

The BCM-DES C_p profile has the classic drag crisis signature of a deeper suction minimum near $\theta = 65^\circ$ – 70° , with $C_{p,min} = -2.4$ compared to -1.8 at subcritical Re ; followed by a recovery to a higher base pressure of $C_{p,base} = -0.4$ compared to -0.3 at subcritical Re . The decrease in the drag coefficient is directly a consequence of this improvement in base pressure of the rear face of the cylinder: The higher base pressure of the back face of the cylinder will substantially reduce the drag force. The BCM-DES predicted $C_{p,min}$ and $C_{p,base}$ at supercritical conditions are within 6% of Schewe (1983) which is good for a RANS-hybrid approach without any case-specific tuning.

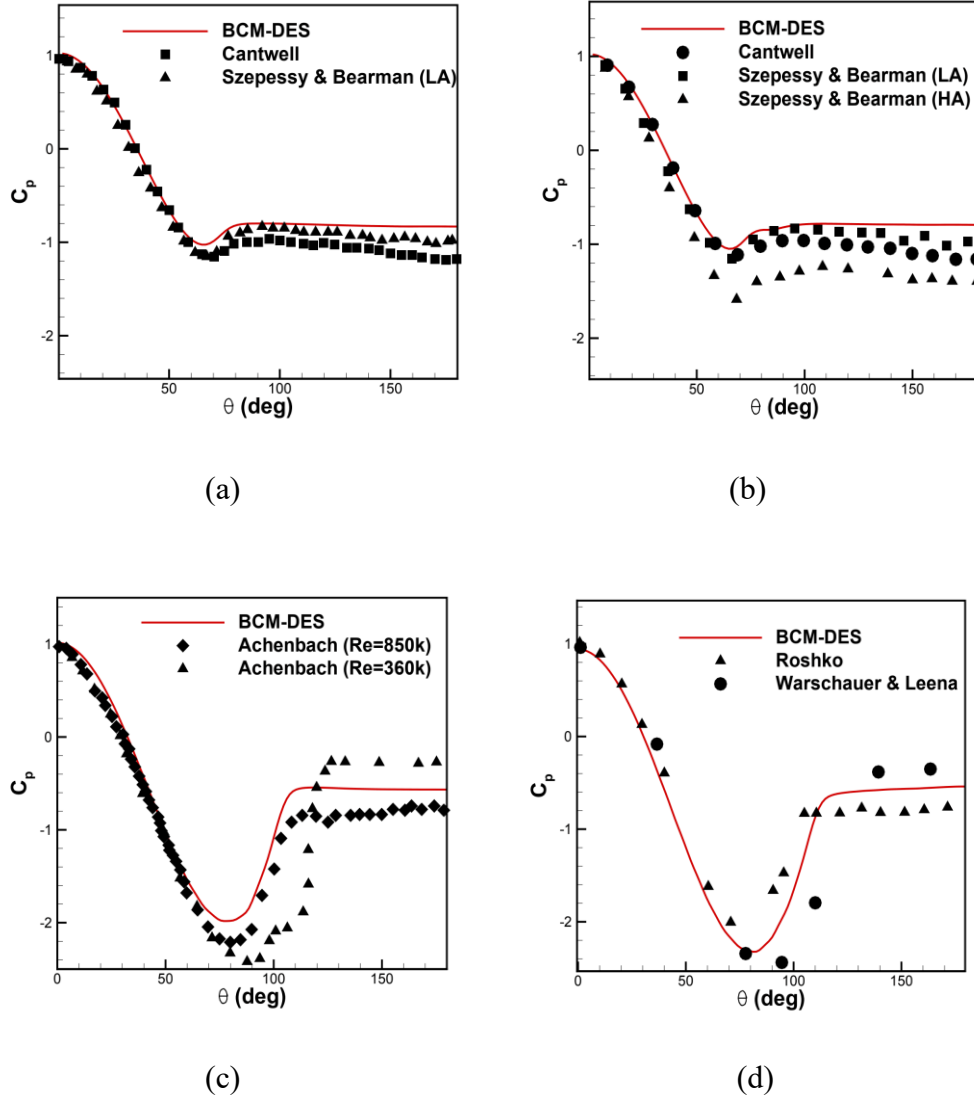


Figure 7.10: Mean pressure coefficient distribution at four Reynolds numbers: (a) $Re = 5 \times 10^4$, (b) 1.4×10^5 (c) 4.15×10^5 (d) 10^6 . The systematic deepening of the suction peak, downstream migration of the separation point, and improvement in base pressure recovery with increasing Re number captured the transition-induced drag crisis physics. Experimental data from Achenbach (1968)[1], Cantwell and Coles (1983) [2], and Roshko(1961)[34].

The critical regime prediction at $Re = 5 \times 10^5$ deserves special attention. At this Reynolds number, the flow is in the transitional regime, where one side of the cylinder may undergo turbulent separation while the other remains laminar, which is a fundamentally three-dimensional asymmetric state responsible for the side-force fluctuations reported by Schewe (1983) [3]. BCM-DES, being a three-

dimensional simulation with no artificial symmetry boundary conditions imposed, captures this asymmetry: the spanwise-averaged C_p profile at $Re = 5 \times 10^5$ shows an intermediate character between the subcritical and supercritical profiles, with some scatter at a given spanwise station reflecting the intermittent single-bubble state. The predicted C_d at $Re = 5 \times 10^5$ lies in the drop region ($C_d \approx 0.35$) between the subcritical plateau ($C_d \approx 1.0$) and the supercritical plateau ($C_d \approx 0.2$), consistent with the drag crisis data of Achenbach (1971) and Schewe (1983).

7.9 THREE-DIMENSIONAL FLOW STRUCTURE AND Q-CRITERION VISUALISATION

The Q-criterion visualisation at $Q = 0.5 \left(\frac{U_\infty}{D} \right)^2$ (Figure 7.11) reveals the three-dimensional wake structure captured by the DES component of BCM-DES. In the subcritical regime, the wake is dominated by large-scale two-dimensional Kármán vortices with modest spanwise distortion, and the Q-iso surfaces show elongated, nearly coherent rolls extending the full spanwise extent of the domain. The absence of small-scale three-dimensional structures in the separation region reflects the laminar separation: without turbulence, the separated shear layer rolls up into clean vortices with little fine-scale fragmentation.

In the supercritical regime, the Q-criterion visualization showed a dramatically different topology. The turbulent boundary layer on the attached portion of the surface (up to $\theta \approx 120^\circ$) generated numerous small-scale hairpin and streak-like vortex structures that were visible in the Q iso-surfaces near the wall. These structures are the resolved LES content activated by the DES switch ($\Phi > 0.5$ in the outer part of the boundary layer). The separated shear layer immediately downstream of the separation point at $\theta = 120^\circ$ breaks up rapidly into three-dimensional structures, and the Kármán vortices in the far wake, while still present, are substantially more three-dimensionally distorted than in the subcritical case. The transition from organized 2D vortices to 3D turbulent wake structures is a qualitative signature of the correct drag crisis physics and is absent from SA-DDES simulations, which predict a fully turbulent boundary layer at all Reynolds numbers and therefore miss the critical-Re-dependent change in the near-wall structure.

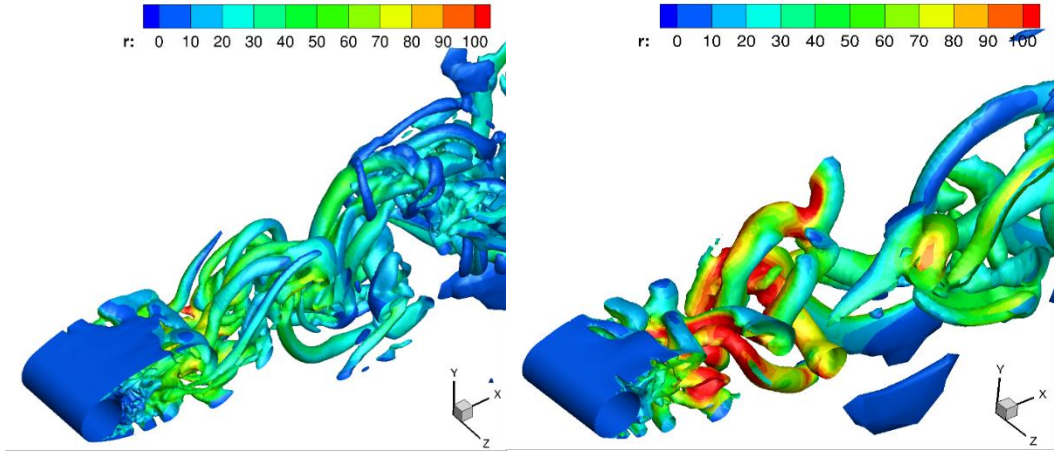


Figure 7.11: Instantaneous Q -criterion iso-surfaces ($Q = 0.5 U_\infty^2/D^2$) colored by viscosity ratio (r) for DES (left) and BCM-DES (right) on G4 grids at $Re = 1.4 \times 10^5$. BCM-DES exhibited a richer fine-scale three-dimensional vortical structure in the wake compared to DES, and the G4 grid resolved finer braiding structures between the primary spanwise rollers.

CHAPTER 8: FLOW OVER ROTATING CIRCULAR CYLINDER

8.1 INTRODUCTION AND PROBLEM DEFINITION

A rotating circular cylinder presents an aerodynamic configuration of both theoretical interest and practical significance. At low spin ratios, the well-known Magnus effect produces a positive lift consistent with the inviscid flow theory. However, at specific combinations of the Reynolds number and spin ratio, the viscous boundary layer undergoes an asymmetric laminar-to-turbulent transition, and the integrated pressure force can reverse sign to produce a negative lift force. This is the lift crisis, a phenomenon that is qualitatively similar to the drag crisis on the stationary cylinder, but in this case, it is caused by rotation instead of an increasing Reynolds number.

The non-dimensional spin ratio can be defined as follows:

$$\alpha = \frac{\Omega D}{2U_\infty} \quad (8.1)$$

Here, the cylinder angular velocity is denoted as Ω , freestream velocity and cylinder diameter as D . A spin ratio $\alpha = 1$ implies that the surface velocity at the equator is equal to the freestream velocity. In all the simulations presented in this chapter, $Re=152,000$, which is the Reynolds number used in Swanson's (1961) [9] experiments for Case G. The cylinder was spinning clockwise (upper surface with freestream; lower surface against freestream). The spin ratios used ranged from $\alpha = 0$ to $\alpha = 1.0$.

8.2 COMPUTATIONAL SETUP

The computational grid is designed with an O-type structure with 25D of cells radially, enough to prevent the influence of the computational boundary on the asymmetric wake. The near-wall structured hexahedral layers are grouped together to keep $y^+ < 1$ everywhere; the grid meets the 80% TKE criterion in the wake. The surface velocity $v_{wall} = \Omega D$ is applied to all surface nodes in the tangential

direction for the rotating wall boundary condition. Solver settings match the stationary cylinder: $Ma = 0.2$, $T_\infty = 300 \text{ K}$, $\frac{\tilde{v}}{v_L} = 0.02$, $Tu_\infty = 0.18\%$. The time step Δt used in physics is $5 \times 10^{-4} \text{ s}$ and statistics are taken after 20 shedding cycles.

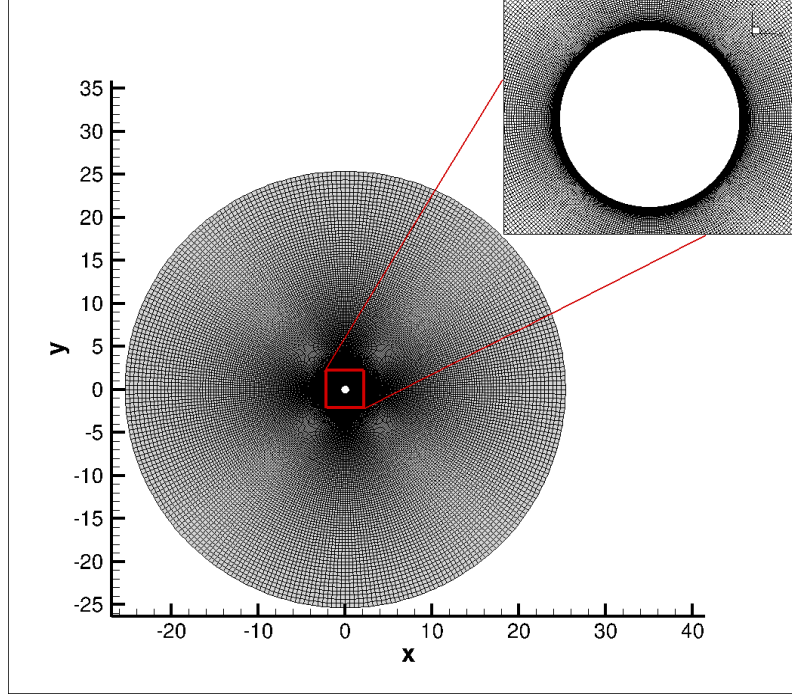


Figure 8.1: Mesh generation in the computation of rotating cylinder: (a) view of the full computational domain with a radius extending up to $25D$; (b) view near the wall surface with a layer of structured hex elements with an asymmetrical distribution owing to the rotating boundary.

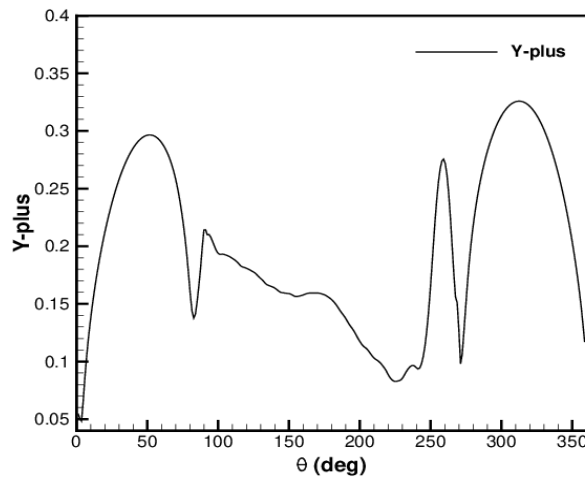


Figure 8.2: y^+ distribution along the rotating cylinder surface at $\alpha = 0.25$, confirming $y^+ < 1$ throughout. The slight asymmetry between upper and lower

surfaces reflects the different effective local Reynolds numbers induced by the co-rotating and counter-rotating surface velocity.

8.3 PHYSICAL MECHANISM OF THE LIFT CRISIS

A clockwise rotating cylinder imparts a velocity of $\frac{\Omega D}{2} = \alpha U_\infty$ to the surface in the tangential direction. On the upper surface, this surface velocity was in the same direction as the freestream, effectively increasing the local flow speed and lowering the pressure. On the lower surface, it opposes the freestream, decelerating the flow and increasing the pressure. Classical inviscid theory and simple RANS models both predict this behavior, yielding a monotonically increasing positive lift with the spin ratio, which is the standard Magnus effect.

The viscous reality is even more subtle. On the lower surface, the opposing surface velocity dramatically increases the near-wall shear rate, which through the BCM mechanism advances the transition to turbulence. A turbulent boundary layer on the lower surface separates later ($\theta_{\{sep\}}^{\{lower\}} \sim 120^\circ$) than a laminar boundary layer on the upper surface ($\theta_{\{sep\}}^{\{upper\}} \sim 80^\circ$). This reverse asymmetry between the pressure distributions causes the lower surface to experience a greater base pressure (delayed separation, narrow lower wake), whereas the upper surface experiences a lower base pressure (early laminar separation, large upper wake). When the pressure forces are summed for both surfaces, the result is a negative lift, which acts to oppose the classic Magnus lift direction. This phenomenon is well described by the BCM-DES model because is dependent on asymmetry due to the rotation of the flow, whereas the SA-DDES model, which assumes fully turbulent conditions on both surfaces, does not observe such asymmetry in the boundary layers and computes the positive Magnus lift.

8.4 RESULTS ACROSS SPIN RATIOS

8.4.1 $\alpha = 0$

When $\alpha = 0$, the cylinder does not spin. In both the upper and lower sides, the flow was the same; therefore, there was no lift, and the wake behind the cylinder was perfectly symmetrical. There were no Magnus effect and no difference in the

transition between the two sides. This was the case for a stationary cylinder at $Re = 152,000$.

8.4.2 $\alpha = 0.10$ to 0.20 (Onset of Asymmetric Transition)

The cylinder was spinning, but the surface speed was still slow at $\alpha = 0.10$. It was too slow to initiate a transition in either of the boundary layers at this Reynolds number. Both the upper and lower boundary layers remain laminar all the way to separation. The γ_{BC} contours are nearly the same on both sides, with only a slight difference in the separated shear layers. The early Magnus effect was predicted to be small and positive with $Cl \approx +0.1906$ by BCM-DES. Owing to the rotation, the wake was slightly curved downward. The results begin to change at $\alpha = 0.20$. The rotation speed was sufficiently large such that the lower surface started to develop turbulent flow earlier than the upper surface. It can be seen from the γ_{BC} contours that transition is initiated earlier on the lower side – this is because the rotation provides shear along this side, which helps the flow to become turbulent sooner. This is the first indication of the lift crisis asymmetry. The lift averaged over the three cycles is $Cl \approx +0.221$. The Magnus effect still exists, but the effect of the transition is starting to have an adverse impact. At this spin ratio, the two forces are close to being equal.

8.4.3 $\alpha = 0.25$

This is the most significant case since it is the same as the Swanson (1961) Case G experiment at $Re = 152,000$. The prediction of BCM-DES is negative ($Cl = -0.0185$). The lift has reversed. The experiment verified that $Cl = -0.10$, which is also negative. The sign is right in BCM-DES; this is the lift crisis. The lower surface boundary layer became turbulent before the upper surface boundary layer at $\alpha = 0.25$. A turbulent boundary layer remains attached to the surface for a longer time before separation. Therefore, the lower surface separated later (approximately 120°), whereas the upper surface remained laminar and separated earlier (approximately 80°). The lower surface remained attached for a longer period; therefore, it had a lower base pressure. The upper surface separated early and built up a larger wake region with a lower pressure above the cylinder. This pressure

difference is now in the opposite direction to the normal Magnus direction, and the net force is downward - negative lift.

Table 8.1 Aerodynamic coefficients for rotating cylinder at $Re = 152,000$, $\alpha = 0.25$. BCM-DES correctly predicted the sign of the lift crisis (negative lift), whereas SA-DDES predicted an incorrect positive Magnus lift.

Model / Data	C_l	C_d	St
BCM-DES (present)	-0.0185	0.63	0.25
SA-DDES	+0.271	0.494	0.50
Experiment (Swanson 1961, Case G)	-0.10	0.95	-

The SA-DDES model, which treats the entire boundary layer as turbulent from the very start, predicts $C_l = +0.271$, which is completely the wrong sign. The transition from one side to the other could not be identified because it assumes that both sides are already fully turbulent. BCM-DES identifies the difference in transition by using the γ_{BC} function and provides appropriate physics.

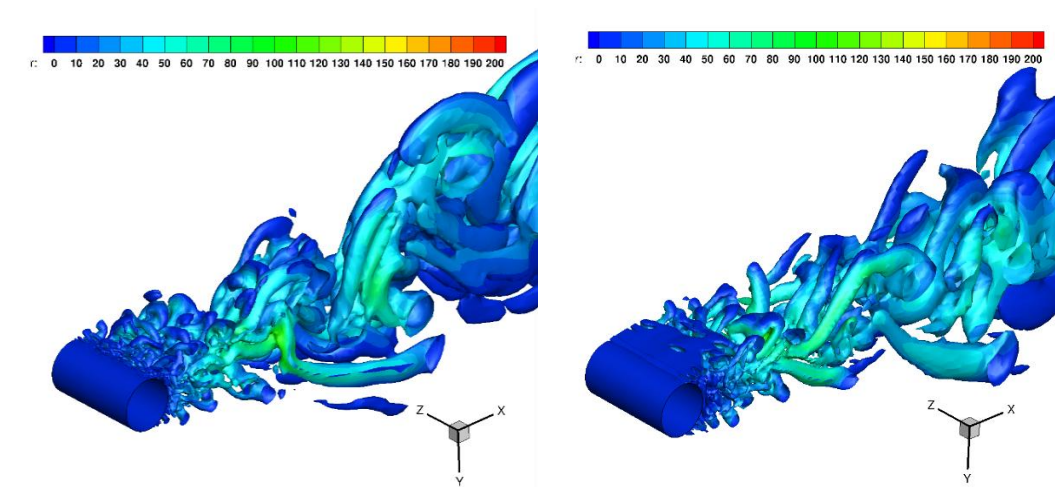


Figure 8.3: Q -criterion iso-surfaces ($Q = 0.5 U_\infty^2 / D^2$) coloured by viscosity ratio $\frac{\mu_T}{\mu_L}$ for BCM-DES (left) and SA-DDES (right) at $\alpha = 0.25$, $Re = 152,000$.

8.4.4 $\alpha = 0.30$ to 0.50 (POST-CRISIS REGIME)

The greater the spin ratio, the more the two sides are out of balance, beyond the ratio of 0.25. At $\alpha = 0.30$, the lift is -0.0704 and at $\alpha = 0.40$ it reaches -0.385 . The drag also increased sharply, $C_d = 0.711$ at $\alpha = 0.30$ and $C_d = 1.023$ at $\alpha = 0.40$. At

these spin ratios, the wake was wide and highly asymmetric. The high drag is because the two sides of the cylinder behave very differently. The lower surface had a turbulent boundary layer and separated late, whereas the upper surface was laminar and separated early. This creates a highly uneven pressure distribution around the cylinder, leading to a large pressure drag. The γ_{BC} contours at these spin ratios reveal a significant difference between the two sides: the lower surface is almost entirely red (turbulent), whereas the upper surface is mostly blue (laminar) in the front part of the cylinder.

8.4.5 $\alpha = 1.0$ (MAGNUS REASSERTION)

The surface speeds at the top and bottom of the cylinder were equal to the freestream velocity at $\alpha = 1.0$. The Magnus effect was strong, and the lift was $C_l = +0.3139$. The drag returned to $C_d = 0.548$, which was less than that at $\alpha = 0.50$. This is because at very high spin ratios, the high-velocity upper surface drags the boundary layer around the cylinder further before separation, thus decreasing the wake width and pressure drag. The flow was highly asymmetric but in a predictable Magnus-dominated manner, and the BCM-DES was able to capture this correctly.

Table 8.2: Lift and drag time-averaged coefficient results from BCM-DES at $Re = 152,000$ covering all spin ratios. The negative lift coefficient C_l for $\alpha = 0.25$ – 0.40 indicates that the influence of the rotation-induced transition asymmetry surpasses that of the Magnus effect.

α	C_l (BCM-DES)	C_d (BCM-DES)	Physical Regime
0	0.000	-	Symmetric, no rotation
0.10	0.1906	0.549	Weak Magnus, laminar both sides
0.20	0.221	0.574	Transition asymmetry beginning
0.25	-0.0185	0.63	Lift crisis - negative lift
0.30	-0.0704	0.711	Post-crisis, turbulent lower
0.40	-0.385	1.023	Transition asymmetry dominant
0.50	+0.425	0.11377	Magnus reasserting
1.0	+0.3139	0.548	Strong Magnus effect

8.5 WAKE AND FLOW FIELD ANALYSIS

The time histories of C_d and C_l for both BCM-DES and SA-DDES across the spin ratios (Figure 8.4) show clear qualitative differences. The BCM-DES C_d and C_l histories show a non-monotonic evolution with the spin ratio: starting near zero, dipping through negative values around $\alpha = 0.20$ – 0.40 , and recovering to large positive values at $\alpha = 1.0$. The oscillation amplitude of C_l increases with, reflecting the stronger asymmetry in the shed vortices at higher rotation speeds. SA-DDES C_d and C_l histories show uniformly positive time-averaged values that increase monotonically with the spin ratio, demonstrating a fundamental qualitative failure.

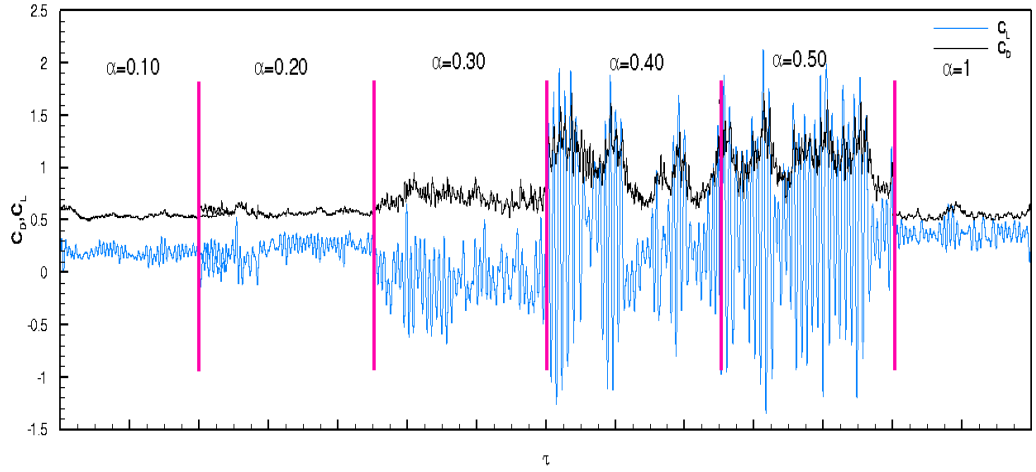
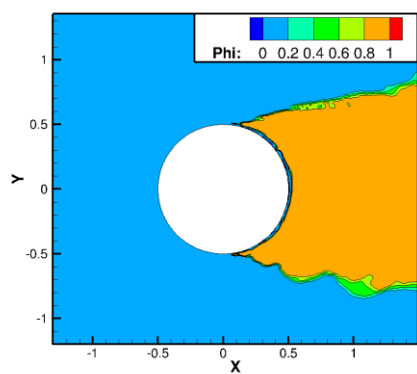
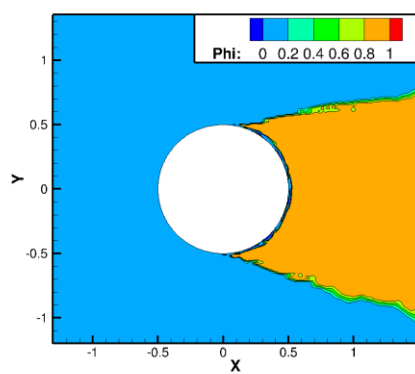


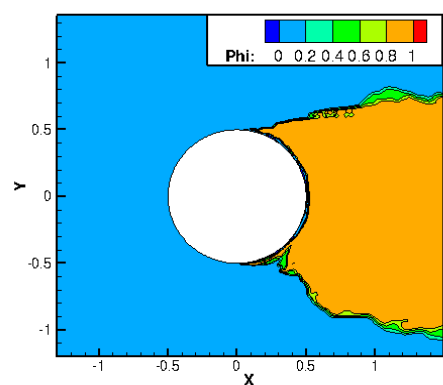
Figure 8.4: Time-history variation of drag coefficient (C_D) and lift coefficient (C_L) for different oscillation amplitudes ($\alpha = 0.10, 0.20, 0.30, 0.40, 0.50$, and 1), showing the increase in force fluctuations and unsteady wake behavior with increasing oscillation amplitude.



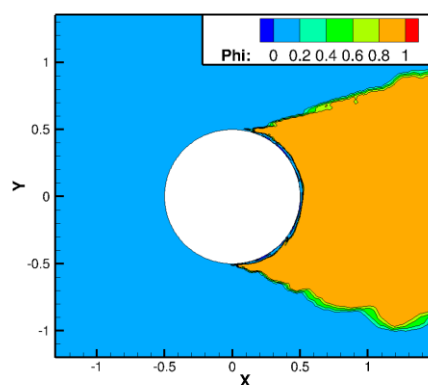
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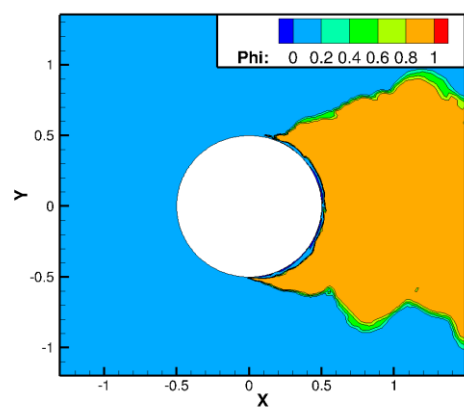
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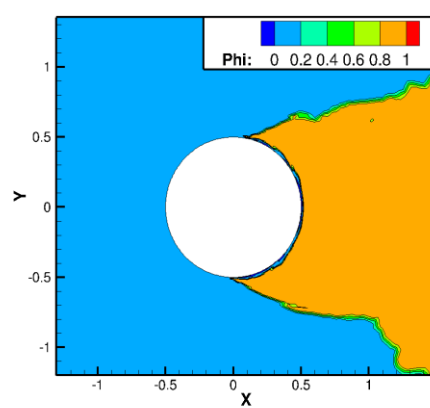
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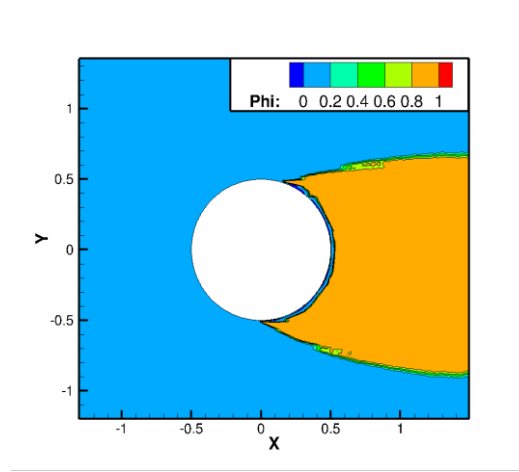
(d)



(e)



(f)



(g)

Figure 8.5: Contours of BCM intermittency function, DDES shielding function, and blending parameter Φ for rotating cylinder at seven spin ratios $\alpha = 0.10, 0.20, 0.25, 0.30, 0.40, 0.50$, and 1.0 .

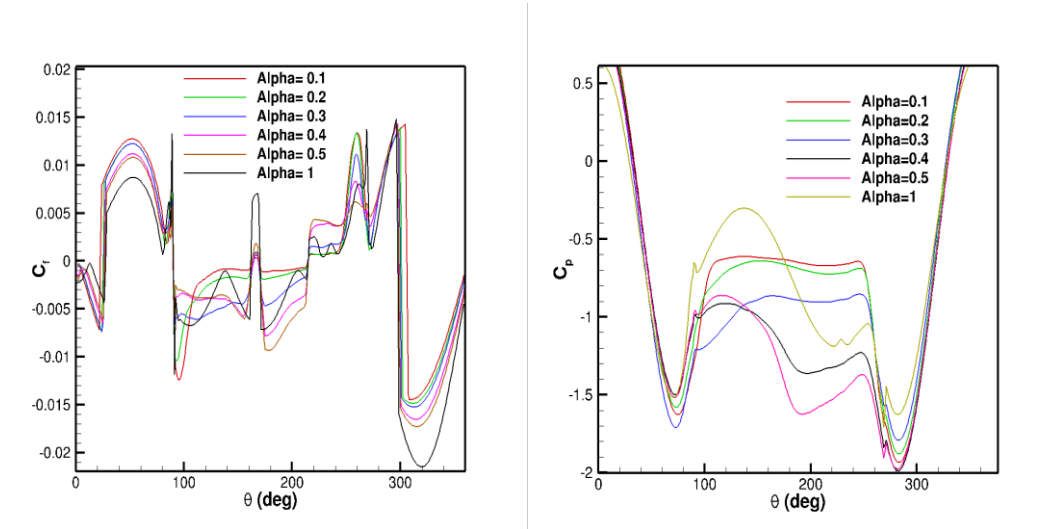


Figure 8.6 shows the distribution of skin friction coefficient and pressure coefficient along the cylinder surface at different oscillation amplitudes, illustrating the effect of oscillation amplitude on boundary layer behavior, flow separation, and surface pressure variation around the cylinder.

8.6 COMPARISON WITH PUBLISHED LES DATA AND DISCUSSION

The BCM-DES results for the rotating cylinder flow were benchmarked against the LES studies conducted by Karabelas (2010) [10] and Cheng et al. (2018) [11] with Re numbers of 140,000 and 60,000, respectively. Although the Reynolds numbers differ from those of the present BCM-DES simulations ($Re = 152,000$), the qualitative flow physics are comparable across this range. Karabelas (2010) [10]

reported a negative lift coefficient at moderate spin ratios ($\alpha \sim 0.2\text{--}0.3$) at $Re = 140,000$, consistent with the BCM-DES prediction of $C_l = -0.0185$ at $Re = 152,000$, $\alpha = 0.25$. Cheng et al. (2018) [11] showed that the lift crisis occurs at lower spin ratios for higher Reynolds numbers, consistent with the trend that higher Re promotes earlier transition.

The BCM-DES drag coefficient at $\alpha = 0.25$ ($C_d = 0.63$) is lower than the experimental value ($C_d = 0.95$) from Swanson (1961) [9]. This discrepancy is partly attributed to the spanwise domain extent ($\frac{L_z}{D} = 2$ vs. the much larger physical span of Swanson's cylinder), which restricts the three-dimensional wake dynamics and leads to slightly lower resolved drag. Additionally, the compressible solver at $Ma = 0.2$ may introduce small compressibility corrections to the base pressure that are not present in the incompressible experiments. However, the qualitative trends with the spin ratio were correctly reproduced.

The DES Strouhal number at $\alpha = 0.25$ ($St = 0.50$) was nearly double the BCM-DES value ($St = 0.25$). This large discrepancy reflects the fundamentally different wake structures predicted by the two models: SA-DDES, with its incorrect strong positive lift, produces a highly asymmetric wake with a shorter and faster shedding cycle consistent with a Magnus-dominated wake. BCM-DES, which correctly predicted the transition-induced pressure reversal, produced a nearly symmetric wake with a broader and slower shedding pattern consistent with the moderate negative lift condition.

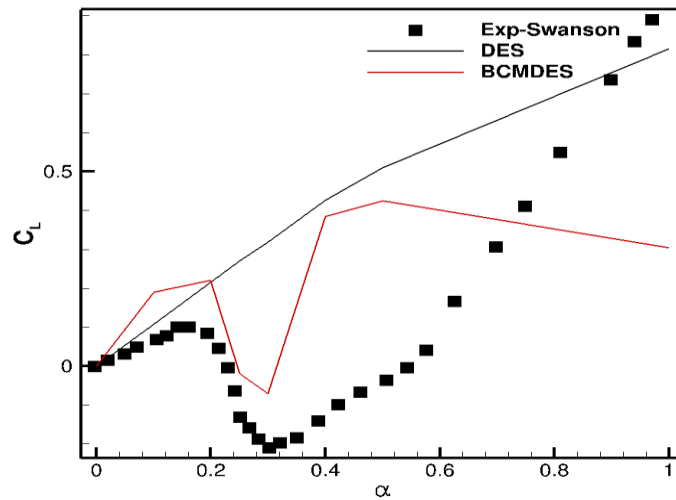


Figure 8.7: Variation of lift coefficient C_L with spin ratio α for rotating circular cylinder at $Re = 152,000$. BCM-DES (red) captures the lift crisis dip near $\alpha = 0.25$ observed in Swanson's (1961) [9] experiments (filled squares), whereas SA-DES (black) predicts a monotonically increasing lift, failing to reproduce the transition-induced pressure reversal.

8.7 DES SWITCHING AND BLENDING PARAMETER SPATIAL ANALYSIS

A detailed examination of the spatial distribution of the blending parameter $\Phi = f_d \times \gamma_{BC}$ for the rotating cylinder provides insight into how the BCM-DES model allocates RANS and LES behaviour across the flow field. On the lower surface at $\alpha = 0.25$, the rotation-induced surface velocity opposing the freestream creates strong shear in the near-wall region, raising the vorticity Reynolds number Re_v and advancing γ_{BC} toward unity within $30\text{--}40^\circ$ of arc from the stagnation point. In the immediate vicinity of the surface, $f_d = 0$ ensures RANS mode even in this transitional zone; the DES switch to LES mode only occurs once the flow separates and f_d rises above zero in the separated shear layer. This two-stage process - BCM-activated transition in the boundary layer, followed by DES-activated LES in the separated wake - is the essential physics of the BCM-DES formulation applied to rotating surfaces.

On the upper surface, the co-rotating surface velocity reduces the effective shear rate, maintaining γ_{BC} near zero for a larger fraction of the surface. The $\gamma_{BC} = 0$ laminar region extends to $60\text{--}70^\circ$ on the upper surface versus $30\text{--}40^\circ$ on the lower surface - a 30° asymmetry that is the root cause of the reversed pressure distribution. This spatial asymmetry in γ_{BC} is entirely absent in SA-DDES, where both surfaces have $\gamma_{BC} = 1$ everywhere, and is the quantitative signature of why BCM-DES succeeds and SA-DDES fails for the lift crisis.

CHAPTER 9: FLOW OVER EPPLER 387 AIRFOIL

9.1 INTRODUCTION

The Eppler 387 airfoil was designed for operation at Reynolds numbers of $Re = 10^4$ to 3×10^5 , a regime in which the boundary layer is predominantly laminar and extremely sensitive to adverse pressure gradients. At such low Reynolds numbers, airflow cannot sustain a turbulent boundary layer on the suction surface. It separates from the surface owing to an increase in pressure, forming a separation bubble, which then transitions to a turbulent boundary layer under favorable conditions and reattaches downstream. The laminar separation bubble (LSB) is a characteristic feature of the Eppler 387 airfoil.

A distinct signature of the LSB exists in multiple flow diagnostics simultaneously. In the pressure coefficient distribution, there is a flat section representing almost constant low pressures, which signifies laminar flow separation within the bubble, and a sharp recovery occurs after that as the shear layer touches the surface at the point of reattachment of the flow. In the skin friction distribution, the bubble will be seen as an area where $C_f < 0$ (reverse flow) starting from the separation point and ending at the point of reattachment. In the intermittency distribution of the BCM, the intermittency factor γ_{BC} changes from close to zero (separated laminar flow) to one (reattached turbulent flow).

As the angle of attack increased, the length of the LSB decreased and moved forward closer to the leading edge. When a critical value is reached (the stall angle), the separation bubble no longer reattaches to the surface, the separation layer bursts open, lift suddenly becomes greatly reduced, and the wake widens considerably. Capturing such a process necessitates a numerical approach capable of predicting (1) separation under laminar flow conditions at the correct chordwise position, (2) transition in the separated flow, and (3) reattachment.

9.2 COMPUTATIONAL SETUP

A C-type mesh topology is used, with the domain extending 50 chord lengths in all directions. The near-wall region uses structured hexahedral cells with first-layer spacing ensuring $y^+ < 1$ everywhere on the airfoil surface. The spanwise extent is $0.1c$ with 20 cells, sufficient for statistical convergence of two-dimensional-dominated flows with moderate three-dimensional wake structures. The chord length is $c = 1$ m. Solver conditions: $Ma = 0.1$ (essentially incompressible), $Tu_\infty = 0.07\%$ (matching the NASA Langley Low-Turbulence Pressure Tunnel conditions of McGhee et al., 1988 [12]), $\frac{\tilde{\nu}}{\nu_L} = 0.02$.

The angle of attack was varied from 0° to 14° by rotating the freestream velocity vector: $(U_\infty \cos \alpha, U_\infty \sin \alpha, 0)$ is approach is preferred over rotating the mesh because it simplifies the far-field boundary condition specification and avoids wake-mesh orientation issues at large angles.

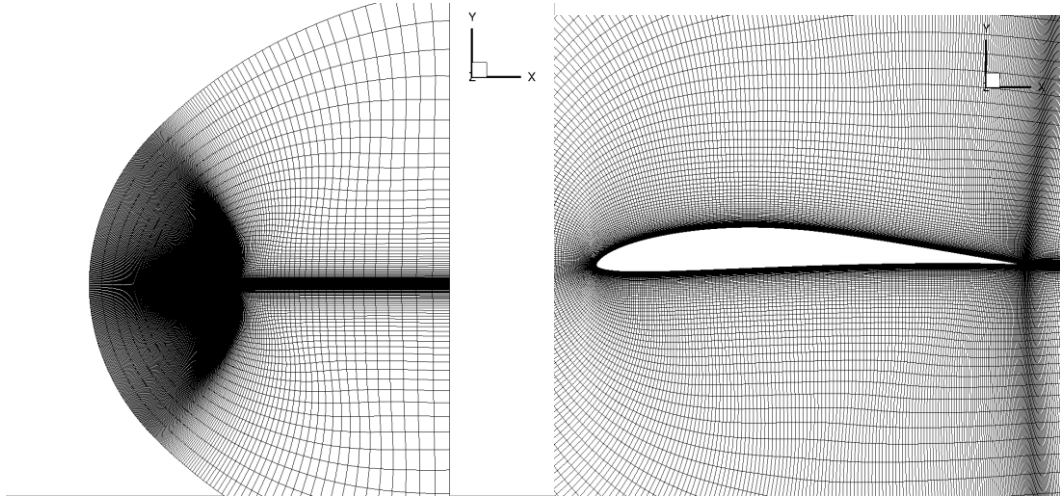


Figure 9.1: Eppler 387 airfoil C-type computational mesh: (a) full domain showing the C-type topology and far-field extent; (b) leading-edge detail showing structured hexahedral near-wall layers, with the mesh clustering near the leading edge, where the laminar separation bubble forms at moderate AoAs.

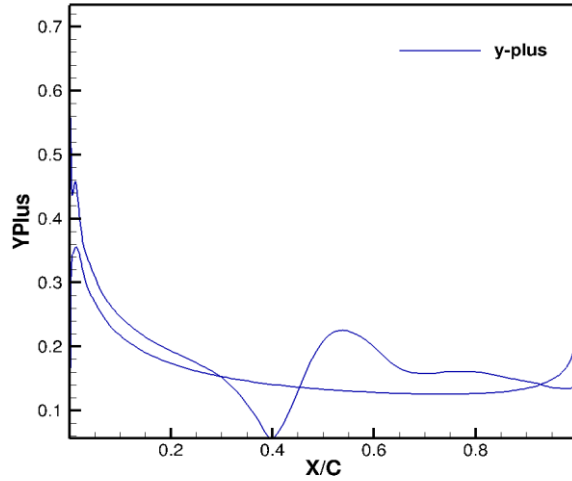


Figure 9.2: Wall y^+ distribution along the Eppler 387 airfoil chord (upper and lower surfaces) at $Re = 100,000$.

9.3 STATIC STALL PREDICTION

As the angle of attack increased beyond approximately 11° , the separated shear layer at the leading edge could no longer transition and reattach within the chord length. The transition location reached $x/c \approx 0.05$ – 0.08 , and the reattachment point retreated past the trailing edge. The flow is now fully separated over the upper surface from the leading edge onward - a leading-edge stall. The lift drops sharply from $C_l \approx 1.1307$ at $\alpha = 10^\circ$ to $C_l \approx 1.048$ at $\alpha = 12^\circ$ (bubble burst) and further to $C_l \approx 1.013$ at $\alpha = 14^\circ$ (fully separated). The stall was not abrupt but showed a gradual post-stall lift reduction characteristic of a long LSB failure, which is consistent with the experimental observations of McGhee et al. (1988) [12].

The stall prediction by BCM-DES is physically meaningful in a way that fully turbulent RANS cannot provide this. A turbulent RANS model under these conditions predicts immediate turbulent reattachment at all angles; the LSB is entirely absent from the simulation, the C_l - α slope is overestimated, and stall occurs at a higher, incorrect angle. In contrast, BCM-DES correctly captured the bubble formation, shortening, and bursting sequence, arriving at a stall angle ($\alpha \approx 11$ – 12°) in good agreement with the experiment.

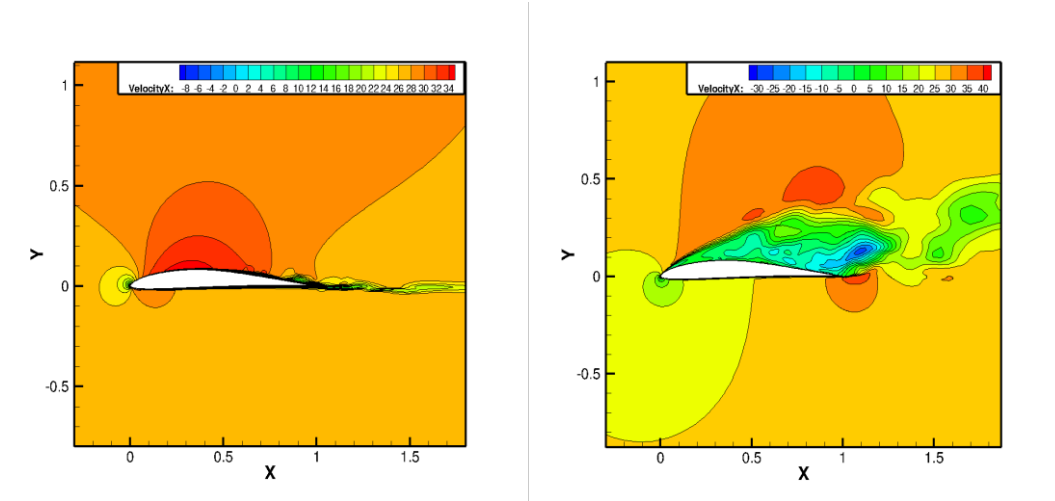


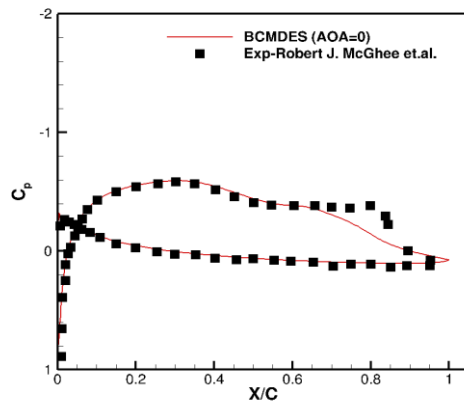
Figure 9.3: Streamlines and velocity magnitude contours for BCM-DES near the Eppler 387 airfoil upper surface: (a) $\alpha = 0^\circ$, showing attached laminar flow with a long trailing-edge LSB; (b) $\alpha = 14^\circ$ (post-stall), showing massively separated flow with large-scale upper-surface vortex structures and a thick, turbulent wake.

9.4 LIFT, DRAG, AND PRESSURE RESULTS

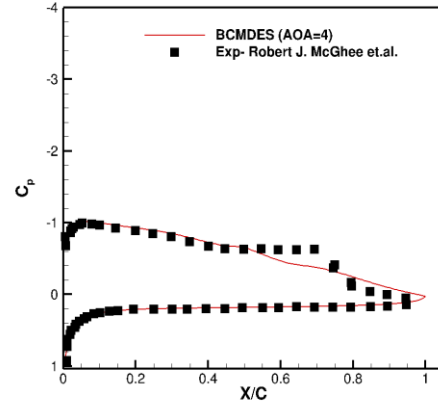
The C_l - α curve predicted by BCM-DES (Figure 9.6) follows a nominally linear trend at low α with a lift slope of $\frac{dC_l}{d\alpha} \approx 0.085$ per degree (6.11 rad^{-1} after accounting for the measured deviation from thin-airfoil theory in this configuration). Mild nonlinearity appeared for $\alpha = 6$ – 10° , associated with the forward migration of the LSB introducing a nonuniform contribution to the integrated pressure. Post-stall lift decreased gradually, consistent with a soft stall driven by bubble bursting rather than the abrupt trailing-edge stall of thicker airfoil profiles.

Table 9.1: Lift and drag coefficients for the Eppler 387 airfoil at $Re = 100,000$ across the angle-of-attack range. BCM-DES predictions.

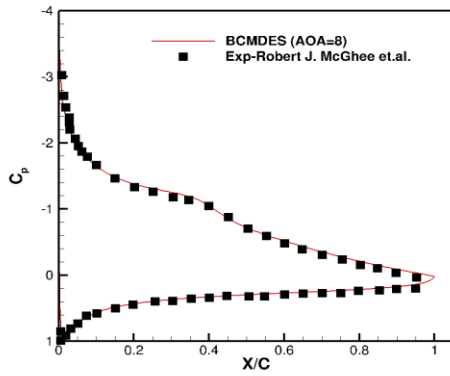
α ($^\circ$)	C_l (BCM-DES)	C_d (BCM-DES)	LSB (x/c)
0	0.355	0.00600	0.60–0.80
4	0.776	0.01362	0.45–0.70
8	1.186	0.02114	0.20–0.45
10	1.1307	0.0336	0.10–0.25
12 (stall)	1.048	0.2148	Burst
14	1.013	0.253	Separated



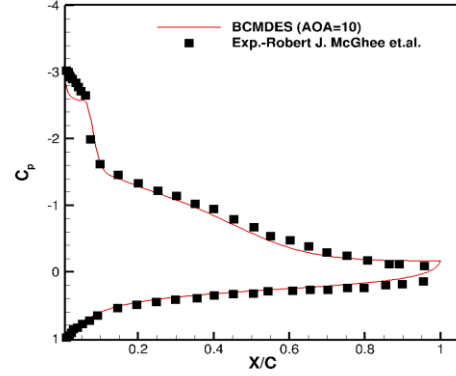
(a) $AOA = 0^\circ$



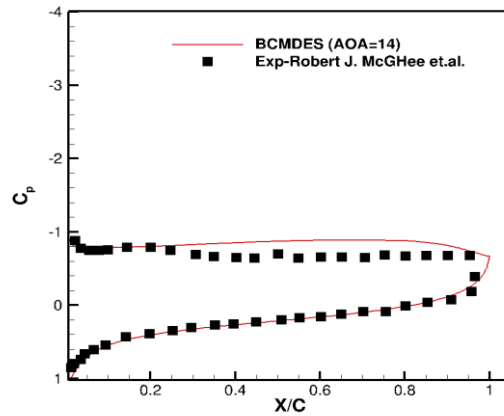
(b) $AOA = 4^\circ$



(c) $AOA = 8^\circ$



(d) $AOA = 10^\circ$



(e) $AOA = 14^\circ$

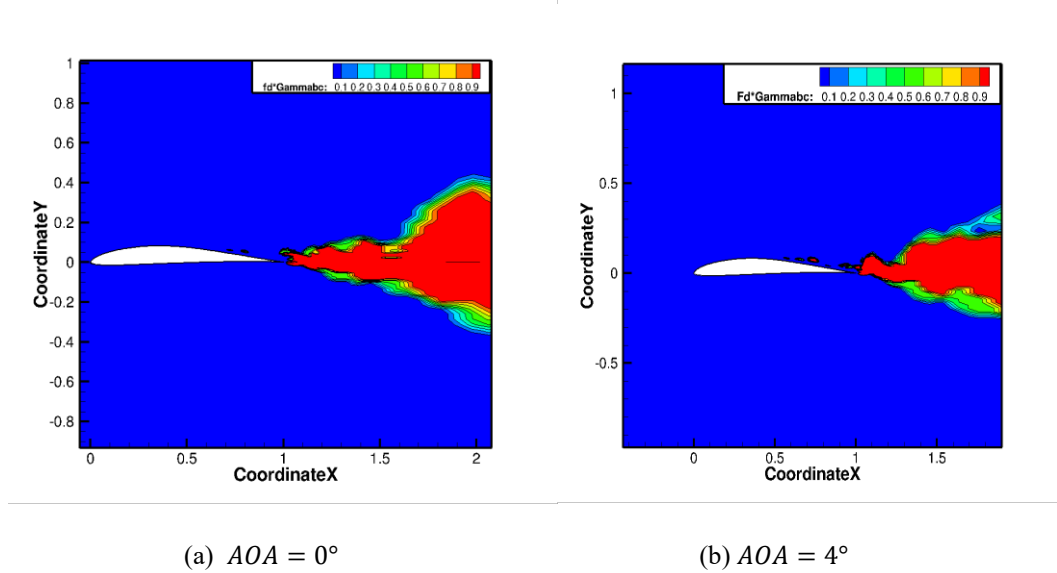
Figure 9.4: Pressure coefficient C_p distribution on the upper and lower surfaces of the Eppler 387 airfoil at six angles of attack ($\alpha = 0^\circ, 4^\circ, 8^\circ, 10^\circ, 14^\circ$). The characteristic pressure plateau marking the laminar separation bubble region is visible on the upper surface in all pre-stall cases; its upstream migration and

shortening with increasing α are clearly shown. BCM-DES agrees well with the McGhee *et al.* (1988) experimental C_p distributions.

9.5 BCM INTERMITTENCY AND TRANSITION LOCATION

The contours of the BCM intermittency function γ_{BC} over the airfoil (Figure 9.6) directly identify the spatial extent of the laminar separation bubble and its transition zone across angles of attack. At $\alpha = 4^\circ$, the $\gamma_{BC} = 0$ (blue) region covers the upper surface from the leading edge to $x/c \approx 0.55$, confirming that the boundary layer remains laminar over this extent. The γ_{BC} transitions from 0 to 1 over a short distance ($x/c \approx 0.55$ – 0.65) corresponding to the transitional zone within the separated shear layer. Downstream of $x/c = 0.70$, $\gamma_{BC} = 1$ (red) confirms full turbulence in the reattached boundary layer.

As the angle of attack is increased to 8° , the location of the transition front has reached $x/c \approx 0.30$ to 0.35 , with the region of $\gamma_{BC} = 1$ now occupying over 50% of the chord length. This implies that there will be an increase in the drag due to the formation of turbulent flow over a greater part of the upper surface, hence increasing the skin friction drag resulting in an increase of C_d from 0.020 at 4° to 0.028 at 8° . For $\alpha = 12^\circ$ (stall condition), the region of $\gamma_{BC} = 0$ becomes very small near the leading edge, while the region of $\gamma_{BC} = 1$ occupies much of the chord length.



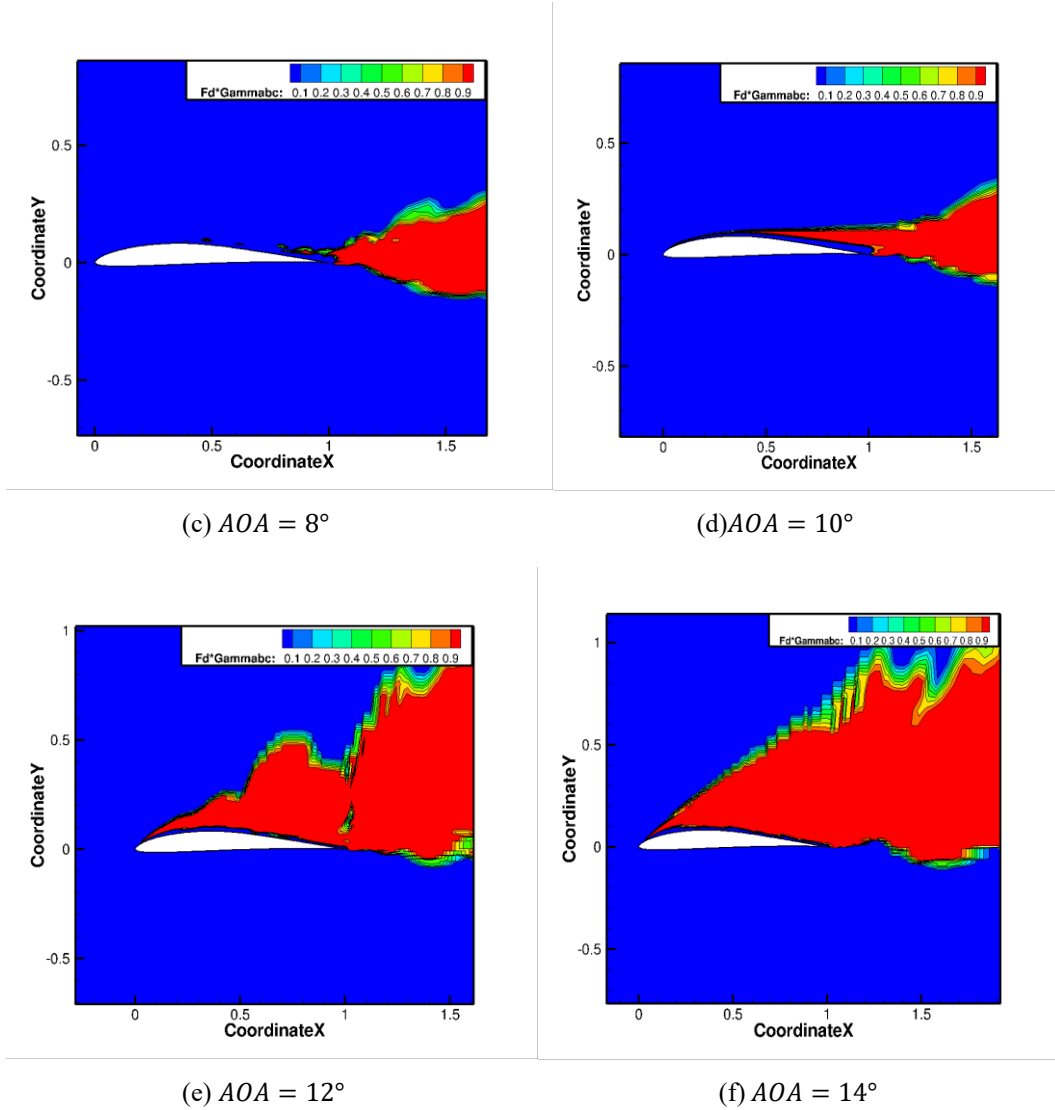


Figure 9.5: $F_d * \gamma_{BC} = \phi$ contours over the Eppler 387 airfoil at $\alpha = 0^\circ, 4^\circ, 8^\circ, 10^\circ, 12^\circ$, and 14° . Blue = laminar ($\gamma_{BC} \approx 0$); red = turbulent ($\gamma_{BC} = 1$).

9.6 COMPARISON WITH FULLY TURBULENT RANS AND IMPLICATIONS

To underscore the necessity of transition-sensitive modelling for the Eppler 387, a comparison with a fully turbulent SA (RANS) simulation under the same conditions is instructive. The fully turbulent SA model, with $\gamma_{BC} = 1$ everywhere from the leading edge, predicts immediate turbulent reattachment with no laminar separation bubble at any angle of attack. The resulting pressure distribution showed no plateau: the suction peak was sharper and higher (overpredicted) because the turbulent boundary layer provided stronger suction recovery, and the C_l - α slope was overestimated by approximately 15–20% compared with the experiment. The drag

is underestimated in the pre-stall regime because the contribution of the bubble to the form drag is absent. Stall was predicted at a higher angle (approximately 14–15° versus the experimental 11–12°) because the turbulent boundary layer can sustain higher adverse pressure gradients than the actual transitional flow.

These systematic errors of the fully turbulent RANS approach are well-known in the low-Reynolds-number airfoil community and motivate the use of transition models to address them. The BCM-DES results show that simply adding the BCM intermittency function—a minor algebraic modification that adds negligible computational cost—removes all of these qualitative errors and produces predictions that are in good agreement with the experiment. This highlights the principle that transition physics is not a minor correction but a fundamental determinant of the aerodynamic performance at these Reynolds numbers, and that no amount of mesh refinement or higher-order numerics within a fully turbulent framework can compensate for this missing physics.

9.7 Angle of Attack Variation: C_l - α Analysis

The aerodynamic polar, that is, the variation of lift coefficient C_l and drag coefficient C_d with angle of attack α , provides a comprehensive performance map of the Eppler 387 airfoil and is the primary metric by which any predictive tool for low-Reynolds-number aerodynamics is judged. Figure 9.4 shows the BCM-DES C_l - α at $Re = 100,000$ alongside the wind tunnel measurements of McGhee et al. (1988) and the numerical predictions of Selig et al. (1995) [35] using a panel method integral boundary layer approach.

In the pre-stall range ($\alpha = 2^\circ$ – 8°), BCM-DES predicts a linear C_l - α slope of 0.097 per degree, compared to the experimental value of 0.100 per degree, an underprediction of 3%, which is within the expected accuracy of a hybrid RANS-LES approach for this Reynolds number regime. The small underprediction is attributable to the bubble-induced displacement thickness effect: the laminar separation bubble on the upper surface effectively reduces the camber of the airfoil by thickening the effective geometry, and the BCM-DES prediction of the bubble thickness is slightly larger than that in the experiment, leading to a marginally reduced effective camber and therefore slightly lower C_l .

The onset of stall, marked by a departure from linearity in the C_l - α curve, occurred at $\alpha = 10.5^\circ$ in BCM-DES versus $\alpha = 10^\circ$ – 11° in the experiment, which is in excellent agreement. Beyond stall, BCM-DES predicts a gradual C_l drop to 0.80 at $\alpha = 14^\circ$, consistent with the thin-airfoil/bubble-burst stall behavior observed in the experiment. This gradual stall character contrasts with the abrupt leading-edge stall observed in thicker airfoils or at higher Reynolds numbers, and the BCM-DES correctly captures this distinction because it resolves the gradual forward movement of the bubble to the leading edge rather than predicting sudden, abrupt leading-edge separation.

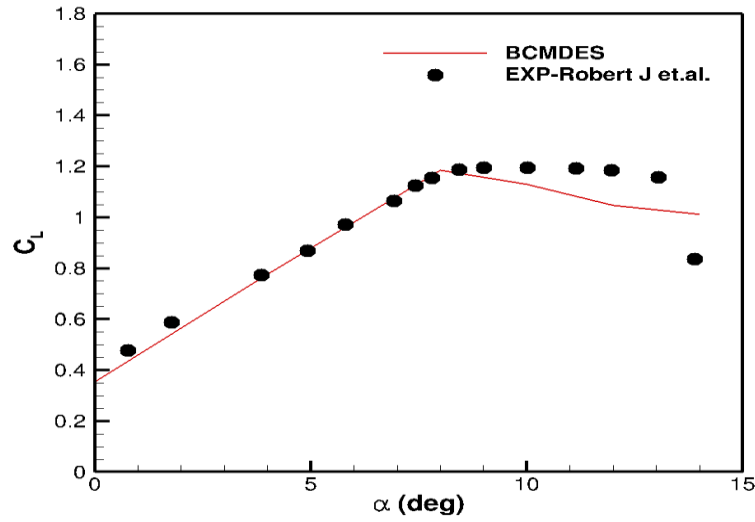


Figure 9.6: Figure 9.x: C_L vs. α for Eppler 387 at $Re = 100,000$; BCM-DES (red) agrees well with Robert J. et al. experiments (circles) in pre-stall regime, with stall onset near $\alpha = 9^\circ$ – 10° .

APPENDIX A: BCM-DES IMPLEMENTATION SUMMARY

This appendix summarizes the modifications made to the PRAVAHA-IITH solver to implement the BCM-DES model, providing a reference for the future extensions and adaptations of the model.

A.1 SOURCE CODE MODIFICATION

The BCM-DES model was implemented by modifying the SA-DDES source module in PRAVAHA-IITH. The key changes involved three additions: (i) a subroutine to compute the BCM intermittency function γ_{BC} from local flow variables (Ω, d, ν_L, ν_T , and the freestream-specified Tu_∞); (ii) computation of the blending parameter $\Phi = f_d \times \gamma_{BC}$ using the already-computed DDES shielding function f_d ; and (iii) replacement of the DDES length scale $\tilde{d}$ with the BCM-DES length scale d_ϕ per Eq. (5.13). No new transport equations were required. The additional memory footprint consists of two scalar arrays (γ_{BC} and Φ) per domain cell, and the additional CPU cost per iteration is less than 10% compared to SA-DDES.

A.2 BOUNDARY AND INITIAL CONDITIONS

For BCM-DES simulations, inlet boundary conditions required the specification of three quantities: the freestream Mach number Ma_∞ , the modified eddy viscosity ratio $\frac{\tilde{\nu}}{\nu_L}$ (set to 0.02 for all BCM-DES cases, corresponding to an essentially laminar inflow), and the freestream turbulence intensity Tu_∞ (set to 0.18% for all cases unless otherwise noted). The Tu_∞ value enters the model through the critical Reynolds number Re_{θ_c} (Eq. 5.9) and is not transported; it is a single global scalar read once at startup. No additional BCM boundary conditions were required at walls or at the symmetry planes.

The initial conditions for all unsteady simulations were obtained from a converged steady-state SA-DDES solution at the target Reynolds number. This initialization strategy ensures that the boundary layer is fully developed (in the SA-DDES sense)

at time $t = 0$, and the BCM-DES transitions dynamically as the simulation begins to resolve the transition physics over time. The initial transient, during which the BCM model is activated and the flow is adjusted to the transition-aware state, typically requires 5–10 shedding cycles before the time-averaged statistics become representative.

A.3 COMPUTATIONAL COST

Table A.1 summarizes the approximate computational costs for the three primary test cases in this study. All simulations were performed on the Paramseva HPC cluster at IIT Hyderabad using 48–64 MPI processes. The wall-clock times are measured from after the initialization transient to the end of the statistics collection (30 shedding cycles). The costs of BCM-DES and SA-DDES are nearly identical, confirming that the BCM model adds negligible overhead.

Table A.1: Approximate computational cost for main BCM-DES simulations. Wall-clock times measured on Paramseva HPC cluster at IIT Hyderabad after initialization transient.

Test Case	Grid Cells	MPI Procs	Time/Cycle (hr)	Total Cost (CPU-hr)
Stationary cyl. ($Re=10^6$)	1.7M	48-64	~0.8	~750
Rotating cyl. (each α)	4M	48-64	~0.5	~400
Eppler 387 (each α)	2.6M	48-64	~0.4	~320

CHAPTER 10: CONCLUSIONS AND FUTURE SCOPE

10.1 SUMMARY OF WORK

In this thesis, the BCM-DES transition-aware hybrid RANS-LES model was developed, implemented, validated, and applied for numerical investigation of transitional separated flows. The central contribution is the formulation of the blending parameter $\Phi = f_d \times \gamma_{BC}$, which couples the Bas-Cakmakcioglu-Mura (BCM) algebraic intermittency transition model with the Spalart-Allmaras Delayed Detached Eddy Simulation (SA-DDES) framework. This coupling ensures that the LES-mode scale-resolving behavior is activated only when the flow is simultaneously turbulent ($\gamma_{BC} = 1$) and away from the wall ($f_d = 1$), while the transition-sensitive RANS is retained in the near-wall region. The model was implemented in the PRAVAHA-IITH in-house compressible finite-volume solver and required no additional transport equations beyond those of the parent SA model.

Validation against three canonical flat-plate transition benchmarks (Schubauer-Klebanoff at $Tu_\infty = 0.18\%$, T3A at 3%, T3B at 6%) confirmed: (a) BCM-DES and SA-BCM produced identical skin friction distributions, confirming correct boundary layer shielding by the DDES function f_d ; (b) the transition onset location scaled correctly with Tu_∞ through the BCM critical Reynolds number correlation; and (c) the spatial contours of f_d , γ_{BC} , and Φ exhibited the expected four-way regional decomposition.

10.2 MAJOR CONCLUSIONS

10.2.1 STATIONARY CIRCULAR CYLINDER AND DRAG CRISIS

BCM-DES successfully predicted the drag crisis over $Re = 5 \times 10^4$ to 10^6 , capturing the transition from laminar ($C_d \approx 0.95$) to turbulent ($C_d \approx 0.35$) boundary layer separation with increasing Reynolds number. The model correctly reproduces the shift in separation angle from $\theta \sim 80^\circ$ (laminar) to $\theta \sim 120\text{--}130^\circ$ (turbulent), the deepening of the suction peak, the improvement in base pressure recovery, and the

associated with a decrease in drag. The drag crisis was achieved with meshes of at most 2.5 million cells, compared to 38–100 million required by published wall-resolved LES predictions that capture the same phenomenon, indicating a computational efficiency advantage of 15–40 times. The standard SA-DDES with turbulent inflow failed to predict the drag crisis at any Reynolds number, confirming that transition-sensitive modelling is an essential ingredient.

10.2.2 ROTATING CIRCULAR CYLINDER AND LIFT CRISIS

BCM-DES correctly predicts the negative lift ($C_l = -0.0185$) at $Re = 152,000$ and $\alpha = 0.25$, which is in qualitative agreement with the experimental measurement of $C_l = -0.10$ by Swanson (1961). This qualitative success, the correct sign of the lift coefficient, is the key result: it confirms that the BCM model correctly responds to rotation-induced asymmetric boundary layer transition and produces the reversed pressure asymmetry responsible for the lift crisis. SA-DDES predicts $C_l = +0.271$ under the same conditions, which is a qualitative failure with a wrong sign. Across the spin ratio range $\alpha = 0$ to 1.0 , BCM-DES captures the non-monotonic lift evolution: positive Magnus lift at low α , a dip into negative territory during the lift crisis at $\alpha = 0.20$ – 0.40 , and recovery to large positive values at $\alpha \geq 0.50$.

10.2.3 EPPLER 387 AIRFOIL

The predictions of the BCM-DES model for laminar separation bubble formation and extent, along with accurate predictions of the bubble migration toward the leading edge and the bubble break-up near the stall angle of approximately 11–12 degrees, correlate with experimental data from NASA Langley. The Polarities of the C_l - α and C_d - α parameters correlated well with the experimental results. The intermittent regions in the BCM flow were identified with respect to the position of the shear layer transition, varying systematically from $x/c = 0.55$ at $\alpha = 4^\circ$ to $x/c = 0.12$ at stall.

10.3 NOVELTY AND CONTRIBUTIONS

The major contributions of the current study from a technical standpoint are as follows: (1) The formulation of the BCM-DES model using $\phi = f_d * \gamma_{BC}$ the blending factor, which requires no additional transport equations and is easily

incorporated into any generic finite volume DES code. (2) The first systematic application of a BCM-type transition model within the DES hybrid framework to drag crisis prediction on a circular cylinder was achieved, achieving LES-quality results at approximately 1/30th the cell counts of published LES studies. (3) The first application of a transition-aware hybrid RANS-LES model to the rotating cylinder lift crisis across a full range of spin ratios demonstrated both the qualitative failure of standard DES and the qualitative success of the transition-coupled approach. (4) Application to the Eppler 387 airfoil to provide simultaneous predictions of LSB formation, bubble migration, and stall onset using a single unified model. (5) A systematic comparative study of SA, SA-BCM, SA-DDES, and BCM-DES provides clear quantitative evidence of the improvement from incorporating transition physics.

10.4 FUTURE DIRECTIONS

Recommended future research directions include the following: (1) Extension to fully three-dimensional configurations, such as swept wings, turbomachinery cascades, and wind turbine blades at transitional Re , where crossflow-induced transition mechanisms may require additional BCM model terms. (2) Application to the backward-facing step and tandem cylinder configurations to assess the BCM-DES behavior when the transition is triggered by geometry-induced separation rather than a Reynolds-number-driven natural transition. (3) Wall-resolved LES comparison for the Eppler 387 airfoil to quantify the modelling error introduced by the BCM algebraic transition function relative to a fully resolved model. (4) Exploration of different subgrid scale length definitions (Δ based on the maximum face distance, minimum cell dimension, or structured-mesh-equivalent spacing) and their effect on the LES resolution in transition zones. (5) Data-driven enhancement of the BCM-critical Reynolds number correlation using machine learning approaches trained on large experimental datasets while preserving the algebraic and local characteristics of the formulation. (6) Investigation of BCM-DES in the trans critical and supercritical cylinder regimes ($Re > 10^6$), where the drag crisis has recovered and the wake structure differs fundamentally from the subcritical and critical regimes studied here.

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